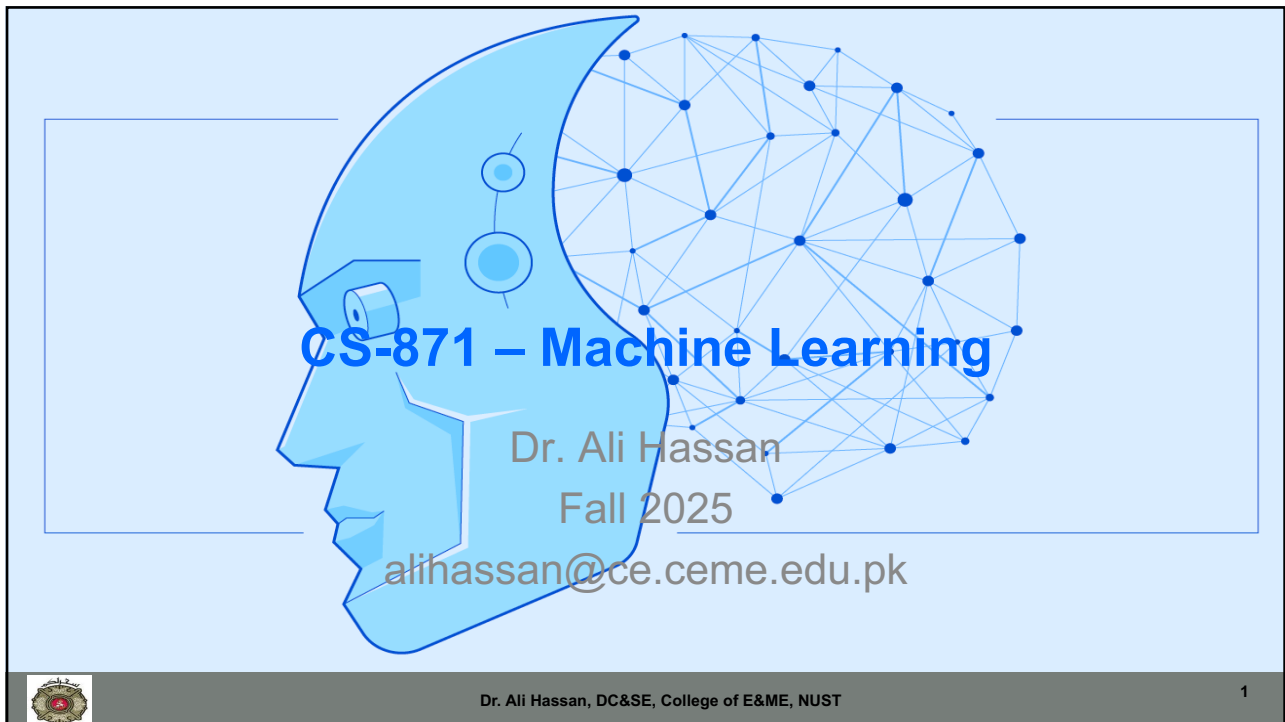




The graphic features a blue profile of a human head facing left. Inside the head, there are two circular nodes connected by lines, and a larger, more complex network of nodes and lines representing a neural network. The text "CS-871 – Machine Learning" is written in blue across the middle of the head. Below it, in brown, is "Dr. Ali Hassan", "Fall 2025", and "alihassan@ce.ceme.edu.pk".

CS-871 – Machine Learning

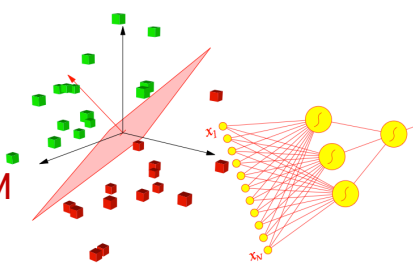
Dr. Ali Hassan
Fall 2025
alihassan@ce.ceme.edu.pk

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1

Announcements

- Assignment 4 -- Project
 - Saturday, 19 December 2025, 11:59 PM
- Assignment 5



The diagram shows a 3D scatter plot with green and red cubes, a red plane, and a neural network diagram with yellow nodes and red connections. Labels x_1 and x_n are present.

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2

Previous Lecture

- Invited lecture on LLMs

Today's Lecture

- Introduction to Deep Learning



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3

3

DEDICATION



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4

4

Dr Abdus Salam

- First Pakistani Nobel Laureate and Theoretical Physicist (29 January 1926 – 21 November 1996)

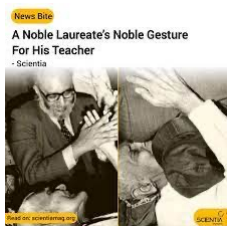


5

5



Dr Abdus Salam receiving his Nobel Award



When Dr Abdus Salam of Pakistan won the Nobel Prize in Physics in 1979, he visited India to meet his teacher, Anilendra Ganguly, 50 years after he had taught him at Lahore



Dr Salam said to him "this medal is a result of your teaching & love of maths that you instilled in me"

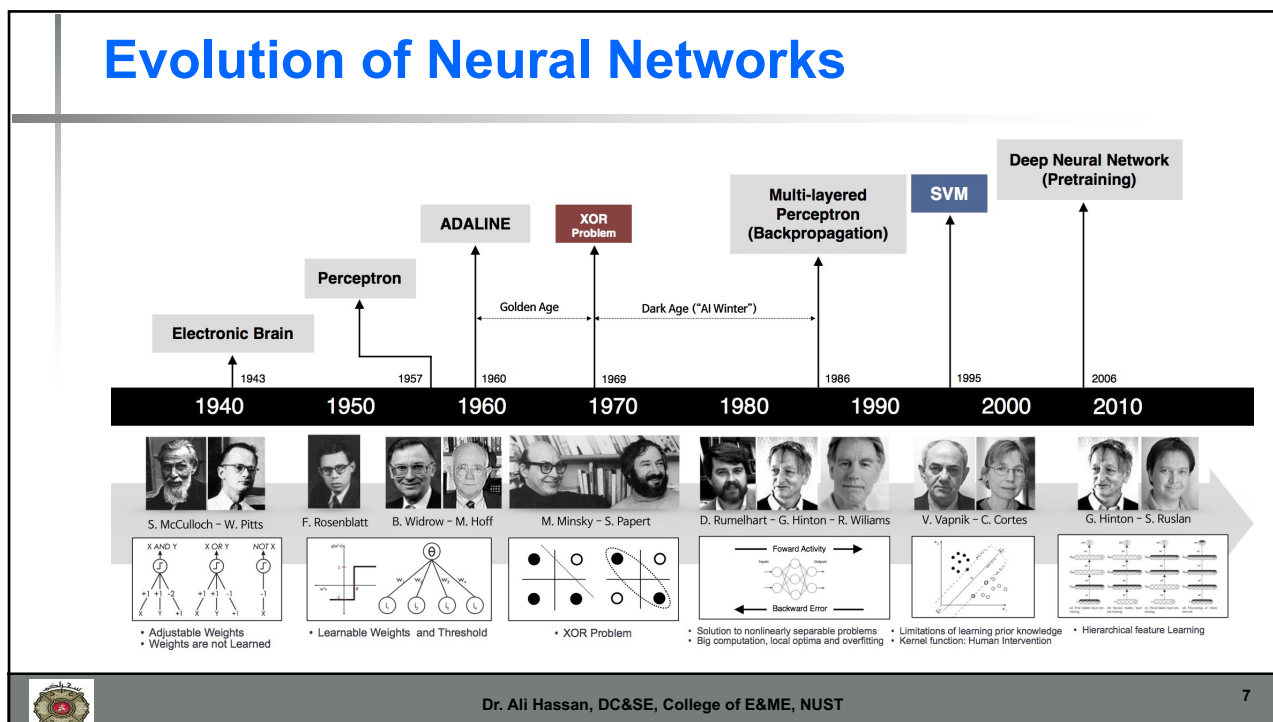
Good education defies boundaries...



6

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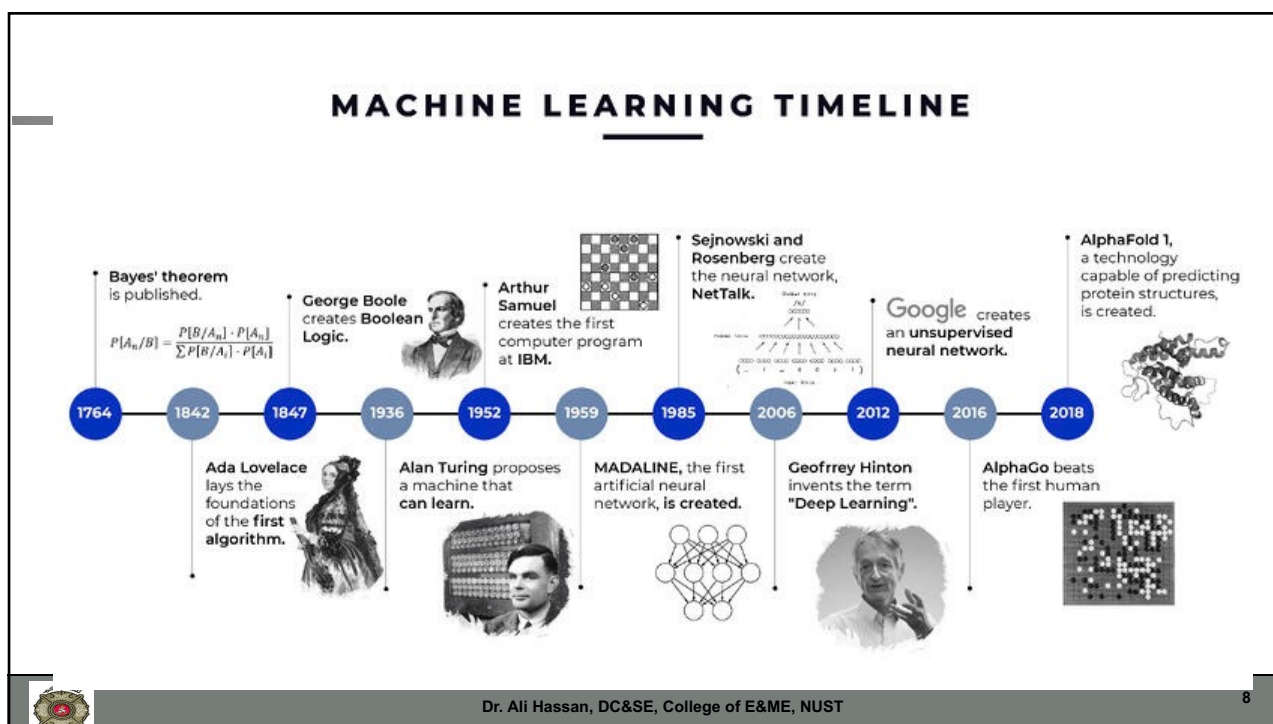
Evolution of Neural Networks



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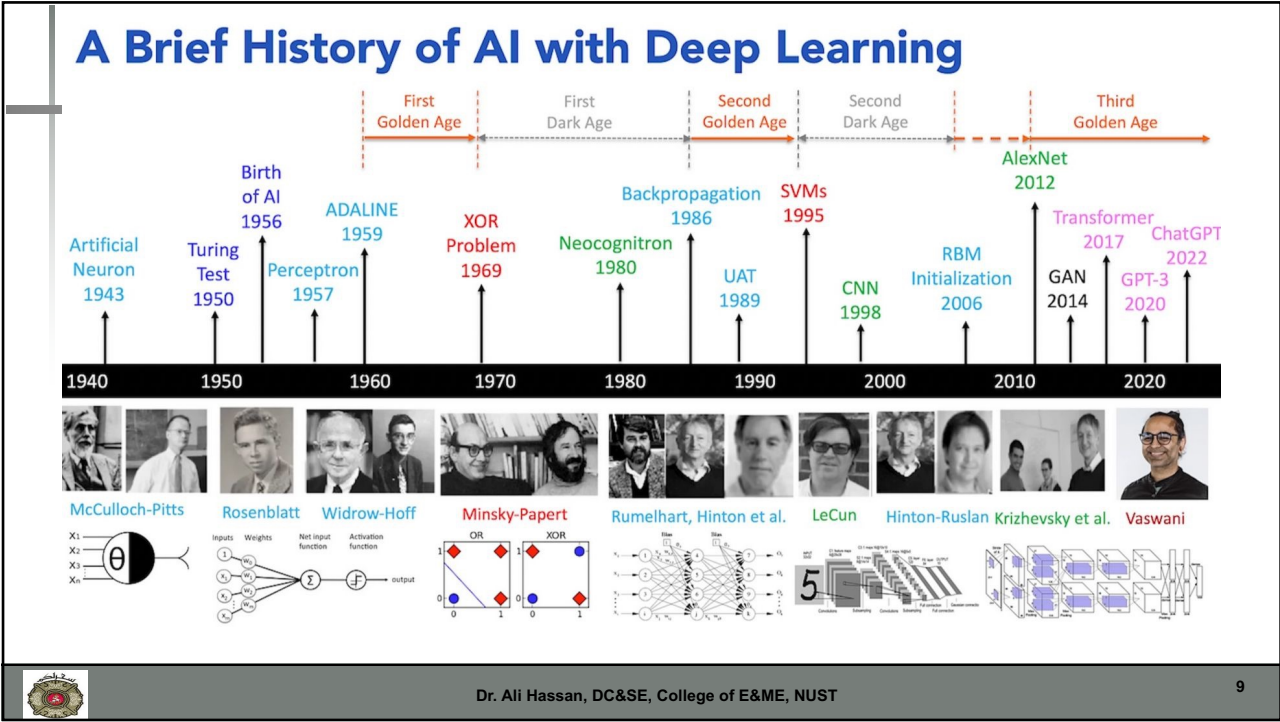
7

MACHINE LEARNING TIMELINE



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8



Deep Learning

INTRODUCTION TO CNN

Simple World



Person



Computer



Alphabet



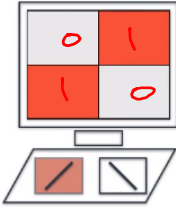
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11

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Keyboard







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12

12

Image recognition software

The diagram illustrates a simple image recognition task. Two computer monitors are shown. The left monitor displays a 2x2 grid where the top-left and bottom-right squares are red, and the top-right and bottom-left squares are gray. A speech bubble next to it contains a backslash '\'. The right monitor displays a 2x2 grid where the top-right and bottom-left squares are red, and the top-left and bottom-right squares are gray. A speech bubble next to it contains a forward slash '/'.

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13

13

The diagram shows the numerical representation of the images from the previous slide. On the left, two 2x2 grids are shown. The top grid, labeled with a backslash '\', has red squares at (1,1) and (2,2), and gray squares at (1,2) and (2,1). The bottom grid, labeled with a forward slash '/', has red squares at (2,1) and (1,2), and gray squares at (1,1) and (2,2). To the right of these are two 1x4 grids. The top 1x4 grid, labeled with a backslash '\', has red squares at positions 1 and 4, and gray squares at positions 2 and 3. The bottom 1x4 grid, labeled with a forward slash '/', has red squares at positions 2 and 3, and gray squares at positions 1 and 4.

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14

14

1

-1

-1

1

-1

1

1

-1

1

+ -1

+ -1

+ 1

= 0

-1

1

1

-1

-1

+ 1

+ 1

+ -1

= 0



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15

15

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-1

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-1

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1

-1

1

x -1

x -1

x 1

= 1

-1

1

1

-1

-1

x 1

x 1

x -1

= 1



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16

16

The diagram illustrates the construction of a 2D array from a 1D array using a 2x2 transformation matrix. The 1D array is represented as a row of four cells: 1 (red), -1 (gray), -1 (gray), and 1 (red). Above each cell is a green sign: $+$, $-$, $-$, and $+$.

The 2D array is represented as a 2x2 grid of cells. The top row contains 1 (red) and -1 (gray), and the bottom row contains -1 (gray) and 1 (red). Above each cell is a green sign: $+$, $-$, $-$, and $+$.

The transformation matrix is shown as a 2x2 grid of cells. The top row contains 1 (red) and -1 (gray), and the bottom row contains -1 (gray) and 1 (red). Above each cell is a green sign: $+$, $-$, $-$, and $+$.

17

The diagram illustrates the calculation of the determinant of a 2x2 matrix using the Laplace expansion method. The matrix is:

$$\begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix}$$

The determinant is calculated as follows:

Expanding along the first row:

$$1 \cdot 1 - (-1) \cdot (-1) = 1 - 1 = 0$$

Expanding along the first column:

$$1 \cdot 1 - (-1) \cdot (-1) = 1 - 1 = 0$$

18

Image Recognition Classifier

+	-
-	+

If positive, “\”
If negative, “/”



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19

An Example



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20

20

The diagram illustrates a transformation between two 2x2 matrices. The top matrix has a red arrow pointing to it from the left, and the bottom matrix has a red arrow pointing to it from the left. The top matrix is:

1	1
-1	1

The bottom matrix is:

-1	-1
1	-1

21

Diagram illustrating two 2x2 matrices:

Matrix 1 (Backslash \):

1	1
-1	1

Matrix 2 (Forward slash /):

-1	-1
1	-1

22

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Diagram illustrating the calculation of the determinant of a 2x2 matrix using the rule of Sarrus.

The matrix is:

$$\begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix}$$

The determinant is calculated as:

$$1 \cdot 1 - (-1) \cdot (-1) = 1 - 1 = 0$$

The diagram shows the expansion along the first row: $1 \cdot 1 - (-1) \cdot (-1) = 1 - 1 = 0$.

The diagram also shows the expansion along the first column: $1 \cdot 1 - (-1) \cdot (-1) = 1 - 1 = 0$.

25

Artificial Intelligence

The diagram illustrates a human-computer interaction in an AI context. On the left, a stick figure represents a human, with a speech bubble containing a 2x2 grid of colored squares (red and grey) and mathematical symbols like backslash and forward slash. On the right, a computer monitor represents the AI, with a speech bubble containing a question mark. The background is a light blue gradient with a dark blue footer bar.

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26

26

Artificial Intelligence

The diagram illustrates a sequence of 16 small 2x2 grids, each containing a 2x2 matrix of '+' and '-' signs. These grids are arranged in two rows of eight, enclosed within a large speech bubble that originates from a computer monitor and keyboard icon on the left. The grids represent different states or configurations in a search space, likely for a puzzle-solving algorithm like A* or Dijkstra's algorithm.

-	-	+	-	-	+	-	-	-	-	+	+	+	-	+	-
-	-	-	-	-	-	+	-	-	+	-	-	+	-	-	+
-	+	-	+	-	-	+	+	+	+	+	-	-	+	+	+
+	-	-	+	+	+	+	-	-	+	+	+	+	+	+	+

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27

27

Artificial Intelligence

The diagram illustrates a sequence of 16 tic-tac-toe boards arranged in two rows of eight. Each board is a 3x3 grid containing '+' and '-' signs. Above each board is a red 'X' indicating a loss, except for the last board which has a green checkmark indicating a win. A speech bubble points from a computer icon to the grid.

Board	Row 1	Row 2	Row 3	Result	
1	-	-	-	-	X
2	+	-	-	-	X
3	-	+	-	-	X
4	-	-	+	-	X
5	-	-	-	+	X
6	+	+	-	-	X
7	+	-	-	+	X
8	+	-	+	-	✓
9	-	+	+	-	✓
10	-	+	-	-	X
11	-	-	+	+	X
12	+	+	+	-	X
13	+	+	-	+	X
14	+	-	+	+	X
15	-	+	+	+	X
16	+	+	+	+	✓

28

Artificial Intelligence

The diagram illustrates a search space for a game, likely Tic-Tac-Toe. It shows a 2x8 grid of small 3x3 grids, each representing a game state. The top row contains 7 incorrect states (marked with red X) and 1 correct state (marked with a green checkmark). The bottom row contains 1 correct state (marked with a green checkmark) and 7 incorrect states (marked with red X).

16 choices

29

Artificial Intelligence

The diagram illustrates a simple decision tree for classification. A computer icon points to a box labeled "so-so". From this box, four arrows point to four different 2x2 grids. Each grid contains "+" and "-" signs in its cells. The first grid is labeled "worse", the second "better", the third "worse", and the fourth "better".

so-so	Grid 1	Grid 2	Grid 3	Grid 4	Label																
so-so	<table border="1"><tr><td>-</td><td>+</td></tr><tr><td>+</td><td>+</td></tr></table>	-	+	+	+	<table border="1"><tr><td>+</td><td>-</td></tr><tr><td>+</td><td>+</td></tr></table>	+	-	+	+	<table border="1"><tr><td>+</td><td>+</td></tr><tr><td>-</td><td>+</td></tr></table>	+	+	-	+	<table border="1"><tr><td>+</td><td>+</td></tr><tr><td>+</td><td>-</td></tr></table>	+	+	+	-	worse better worse better
	-	+																			
	+	+																			
	+	-																			
+	+																				
+	+																				
-	+																				
+	+																				
+	-																				

30

Artificial Intelligence

SO-SO

worse
better!
worse

31

31

Artificial Intelligence

Real weights are not just +1 and -1

0.5	1.2
0.7	1.0

,

0.9	-0.7
-0.6	1.1

, etc...

Too many possible option

32

32

Gradient Descent

The diagram illustrates the iterative process of Gradient Descent. It shows a sequence of weight matrices, each represented as a 2x2 grid. The sequence starts with the initial matrix $\begin{bmatrix} 0.5 & 1.2 \\ 0.7 & 1.0 \end{bmatrix}$ and proceeds through several steps, with each subsequent matrix being a small step closer to the optimal solution. The final matrix shown is $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. A red arrow on the right side of the diagram points downwards, indicating a decrease in the error function, from 'Lots of errors' at the top to 'Few errors' at the bottom.

0.5	1.2
0.7	1.0

0.6	0.9
0.5	1.1

0.7	0.4
0.3	1.2

0.9	-0.2
-0.2	1.1

0.9	-0.7
-0.6	1.1

1	-1
-1	1



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33

33

Gradient Descent

Derivatives

The diagram illustrates the iterative process of Gradient Descent, similar to the one on slide 33. It shows a sequence of weight matrices, each represented as a 2x2 grid. The sequence starts with the initial matrix $\begin{bmatrix} 0.5 & 1.2 \\ 0.7 & 1.0 \end{bmatrix}$ and proceeds through several steps, with each subsequent matrix being a small step closer to the optimal solution. The final matrix shown is $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, which is circled in green. A red arrow on the right side of the diagram points downwards, indicating a decrease in the error function, from 'Lots of errors' at the top to 'Few errors' at the bottom.

0.5	1.2
0.7	1.0

0.6	0.9
0.5	1.1

0.7	0.4
0.3	1.2

0.9	-0.2
-0.2	1.1

0.9	-0.7
-0.6	1.1

1	-1
-1	1



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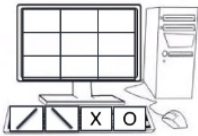
34

34

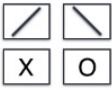
Slightly More Complex World



Person



Computer



Alphabet



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35

35

1	-1	-1
-1	1	-1
-1	-1	1

\

1	-1	1
-1	1	-1
1	-1	1

X

-1	-1	1
-1	1	-1
1	-1	-1

/

-1	1	-1
1	-1	1
-1	1	-1

O



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36

36

1	-1	1
-1	1	-1
1	-1	1

This will be very difficult; we draw on our previous knowledge



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37

37

Previous Knowledge

1	-1
-1	1

-1	1
1	-1



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38

38

1-11

-11-1

1-11

-11-1

1-11

-11-1

-1-11

-11-1

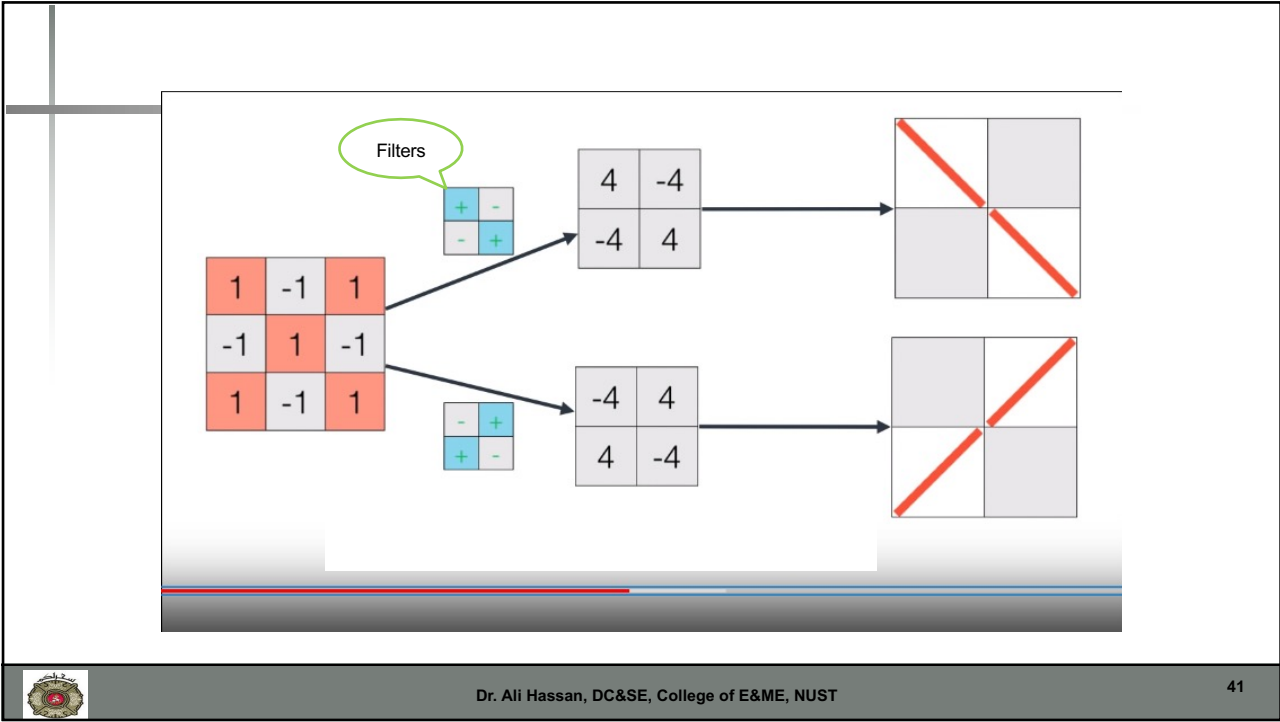
1-11

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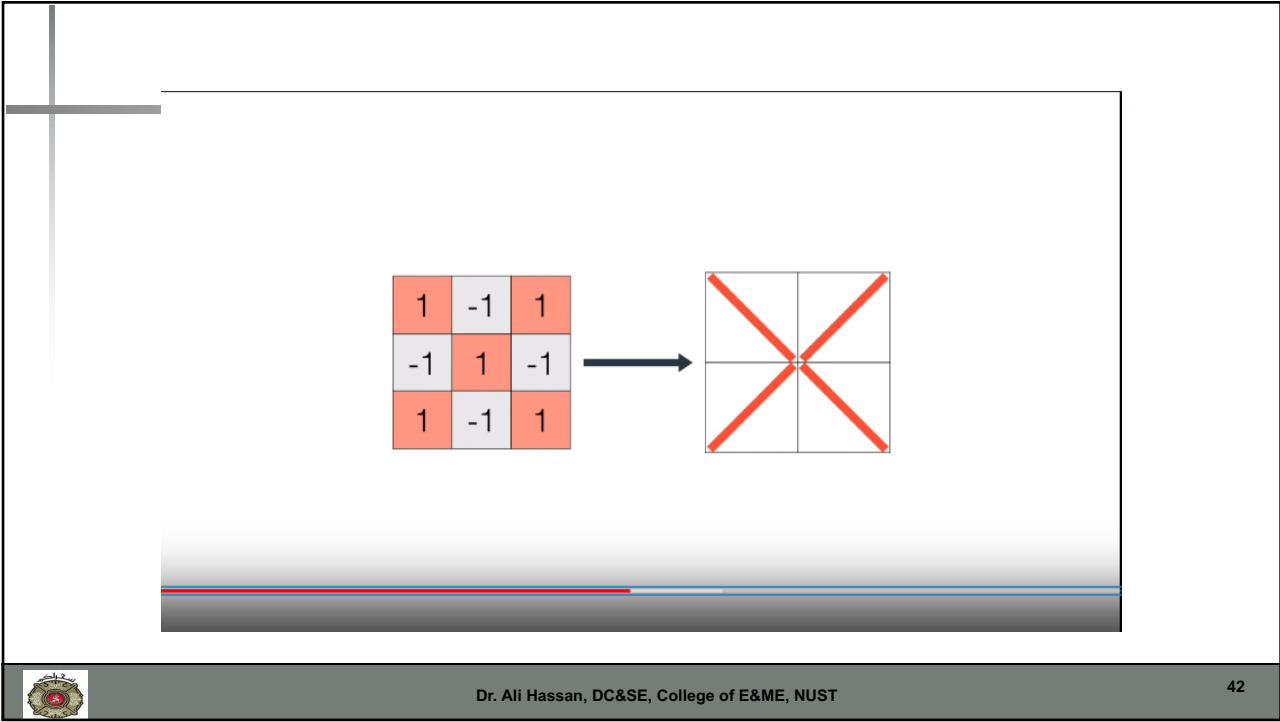
-11-1

-1-11

<



41



42

Diagram illustrating a transformation process:

Initial 3x3 matrix:

1	-1	-1
-1	1	-1
-1	-1	1

The center element (1) is highlighted with a green box.

Two arrows point from the green box to two 2x2 matrices:

Top 2x2 matrix:

4	-2
-2	4

Bottom 2x2 matrix:

-4	2
2	-4

Each 2x2 matrix is then transformed into a 2x2 matrix with a red diagonal line.

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43

Diagram illustrating a transformation process:

Initial 3x3 matrix:

1	-1	-1
-1	1	-1
-1	-1	1

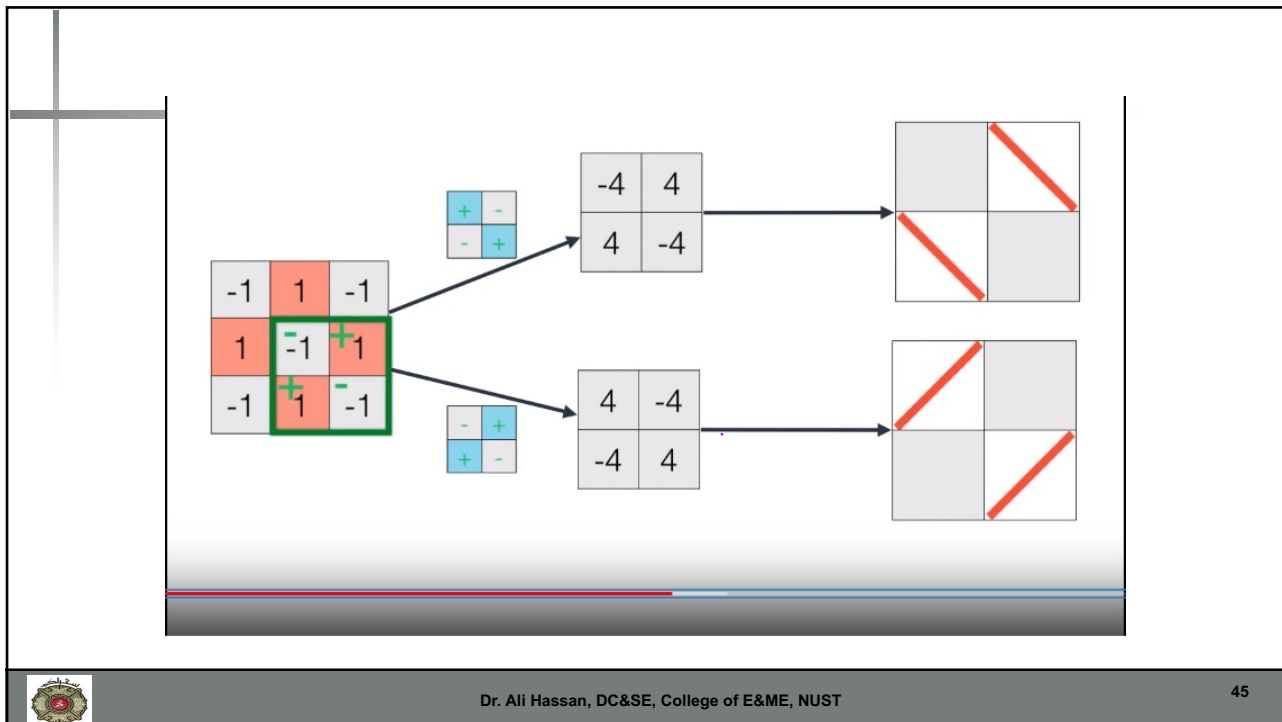
The center element (1) is highlighted with a green box.

An arrow points from the green box to a 2x2 matrix with a red diagonal line.

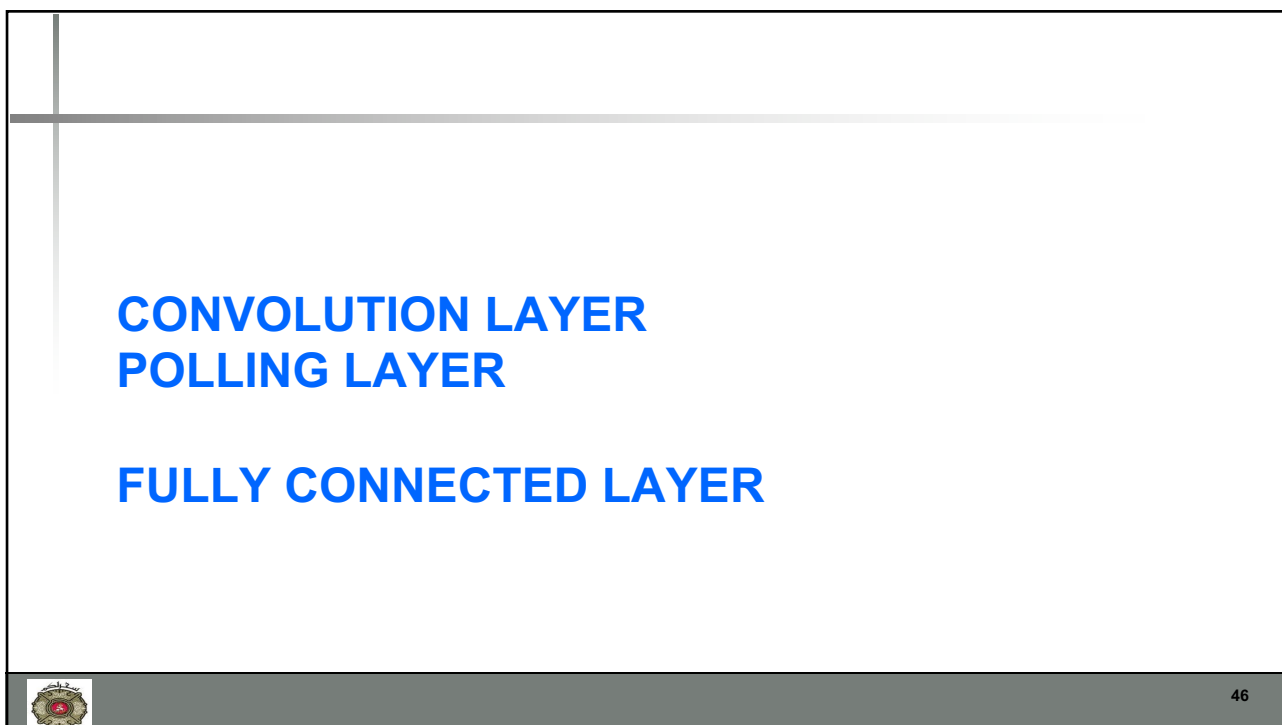
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44

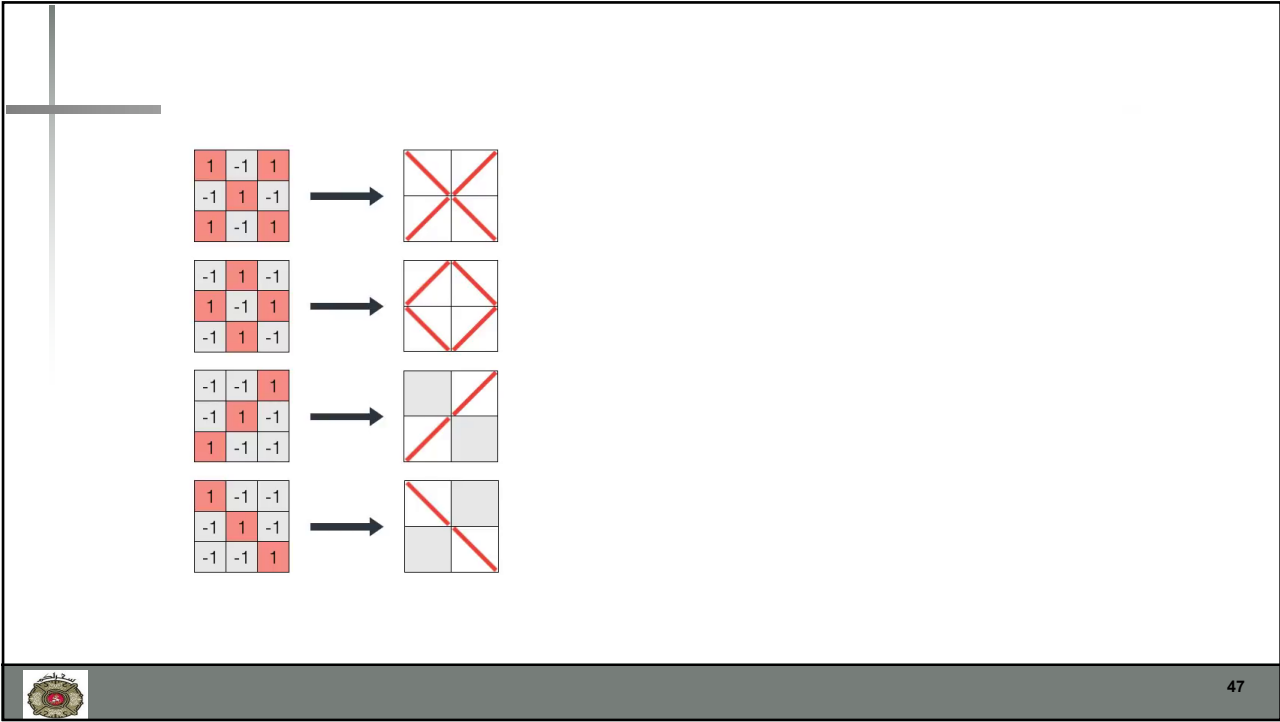
44



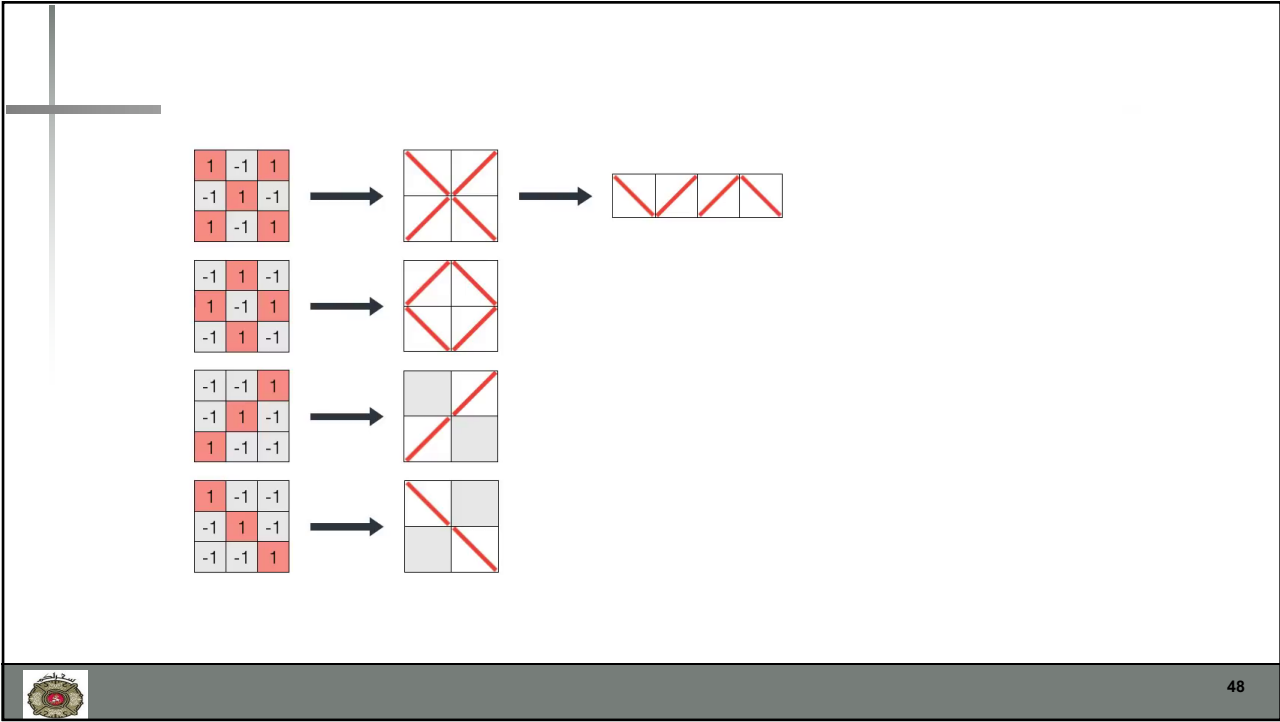
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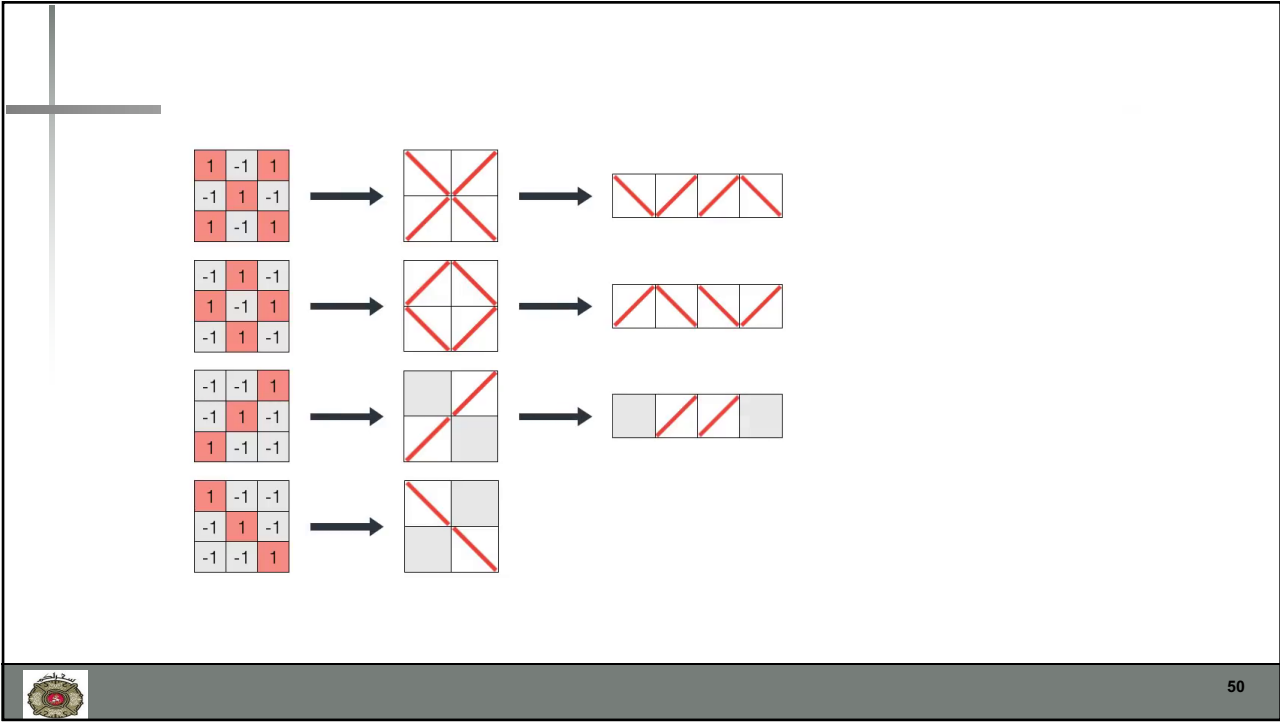
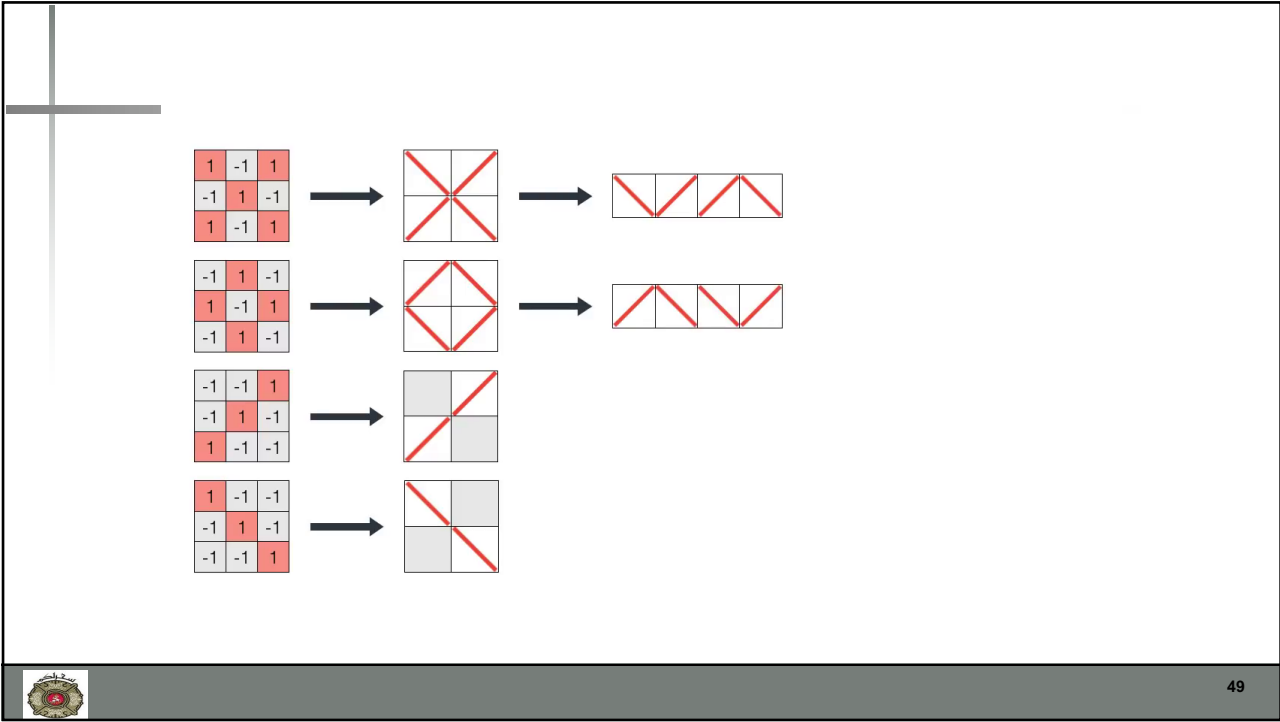
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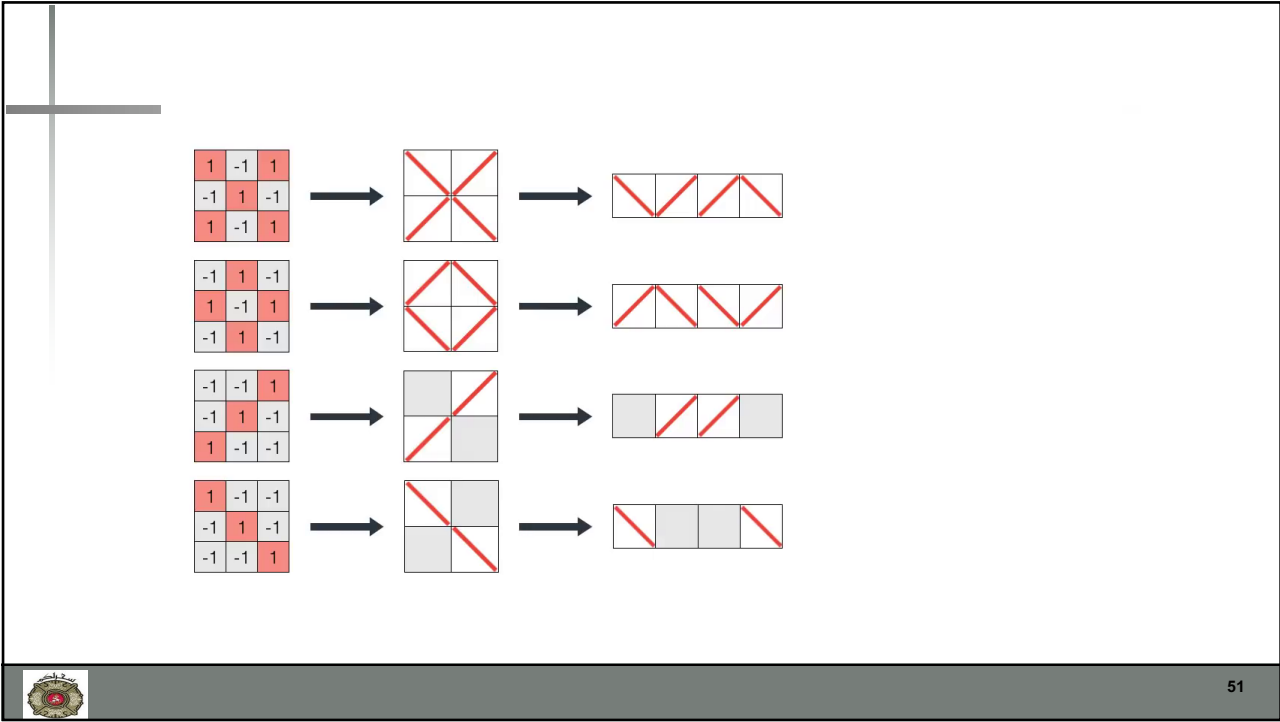


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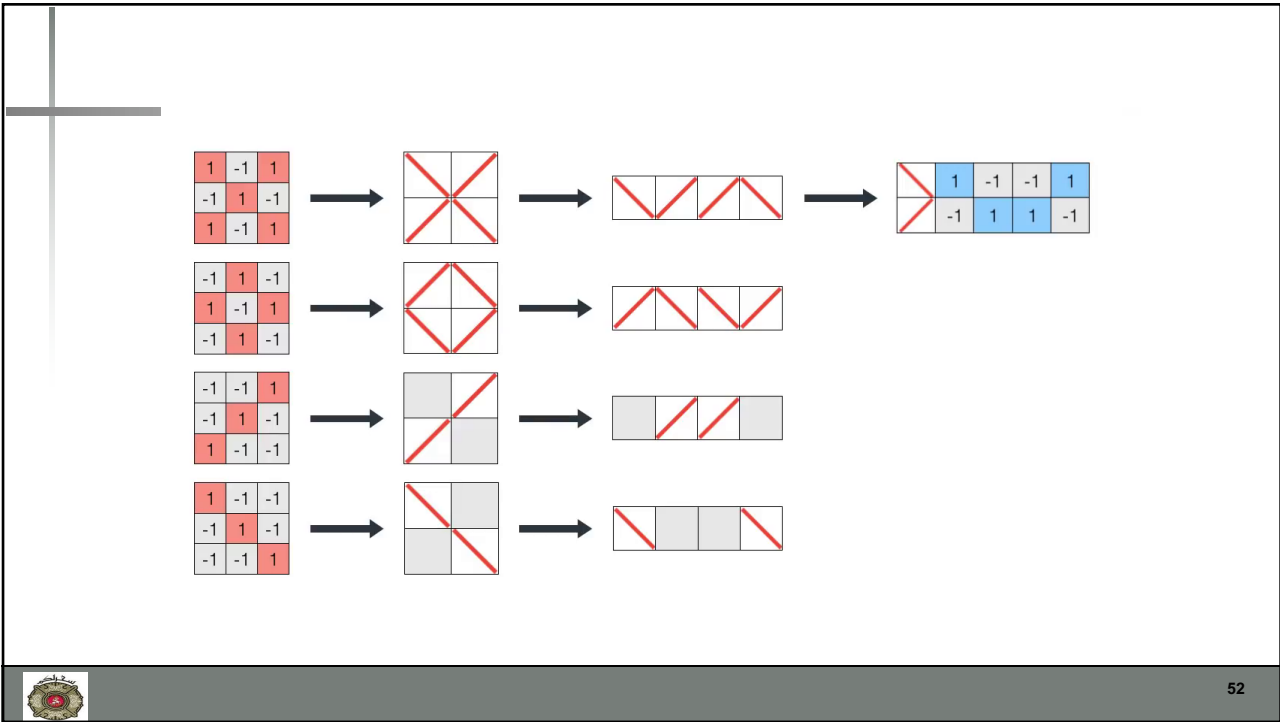


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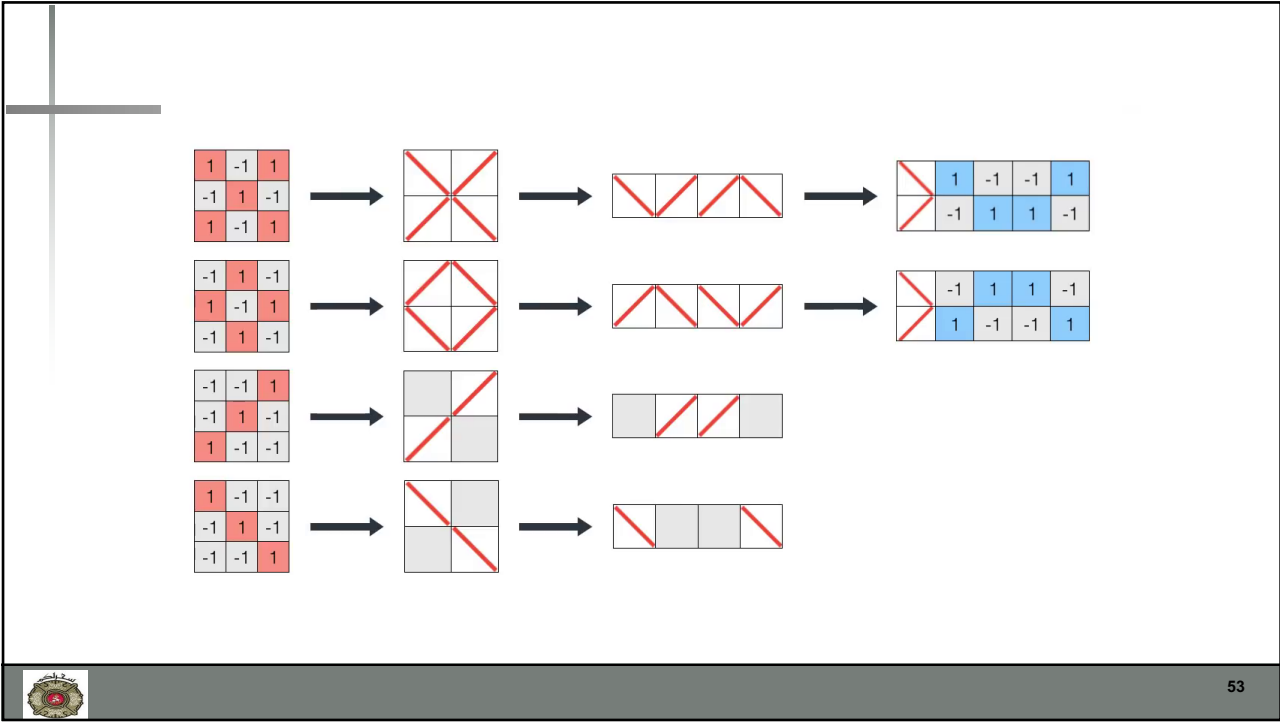




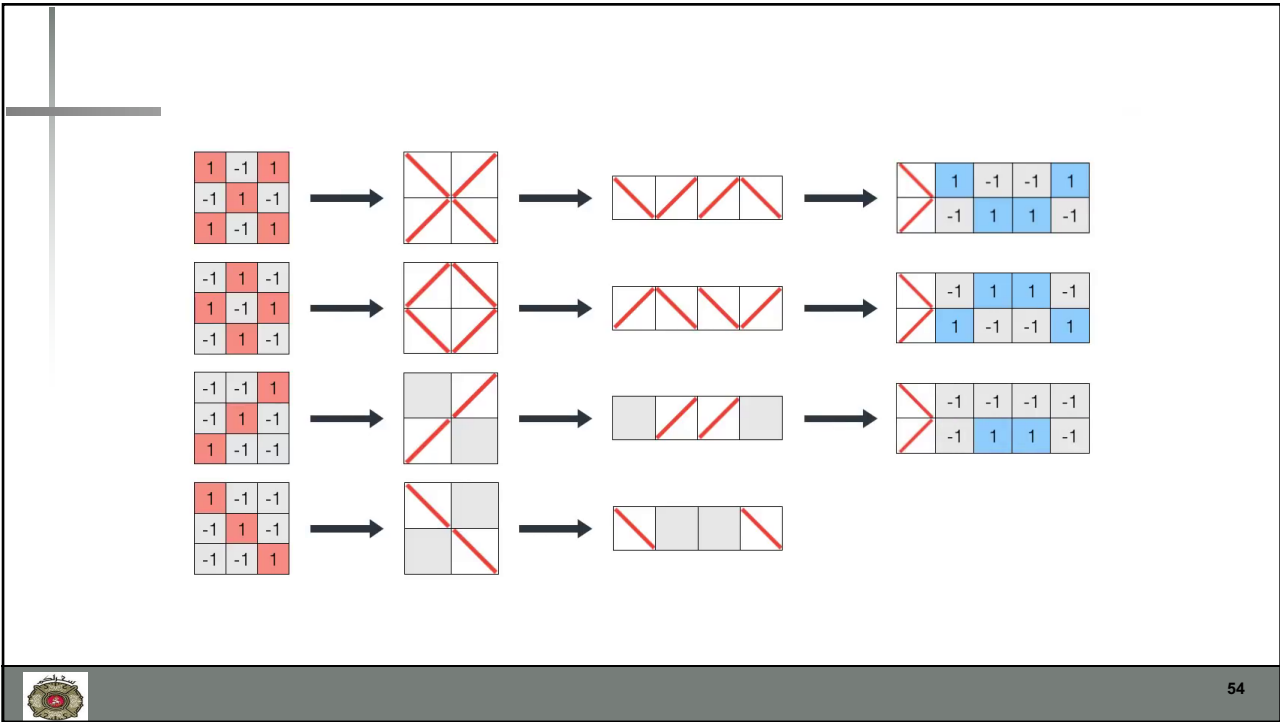
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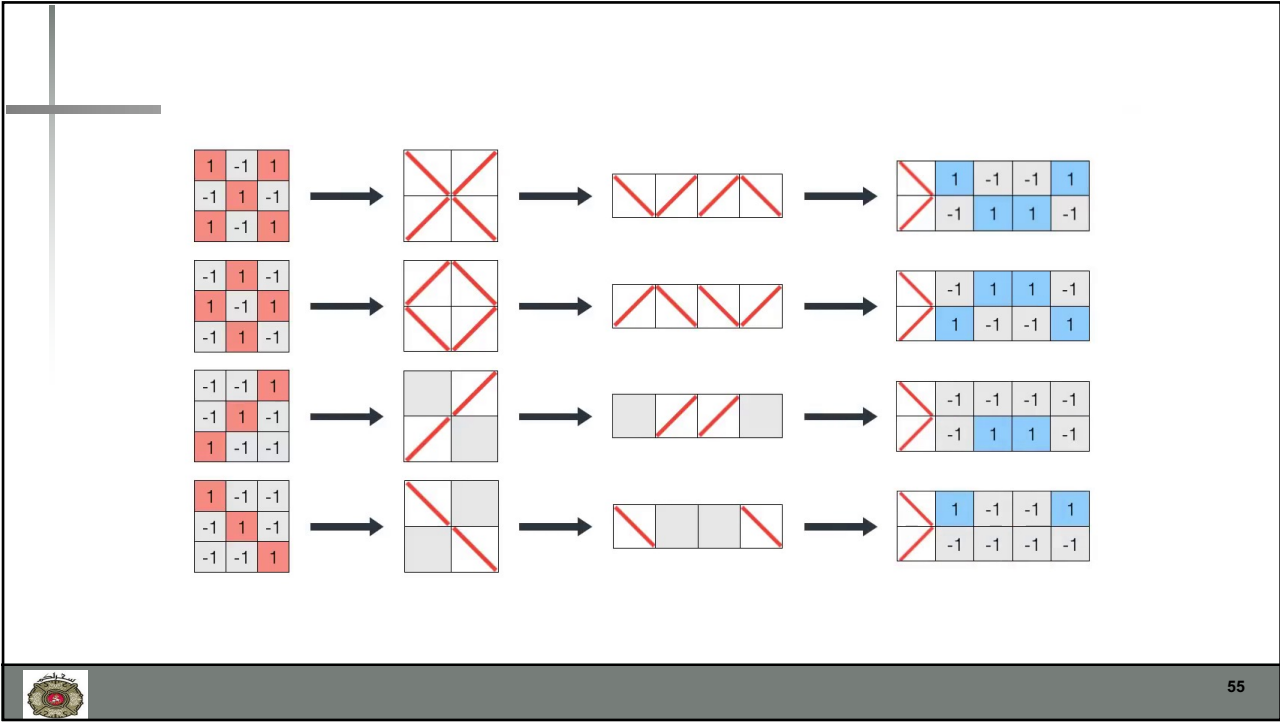
52



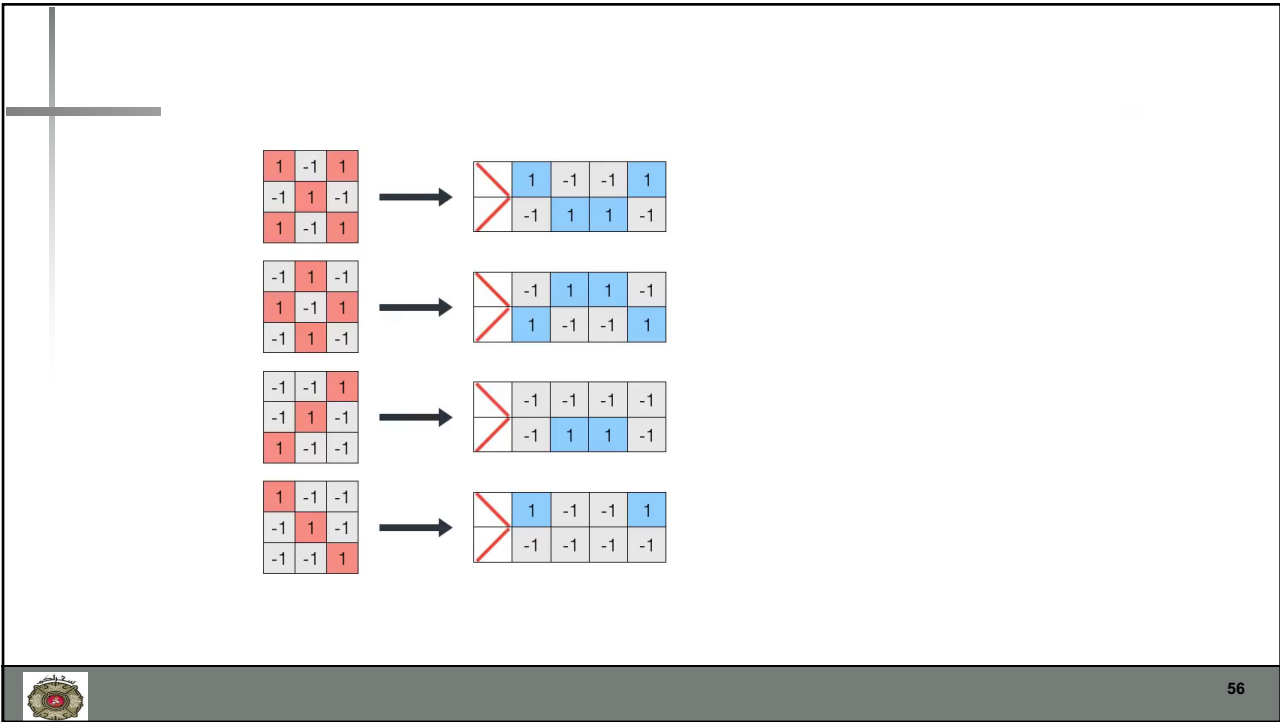
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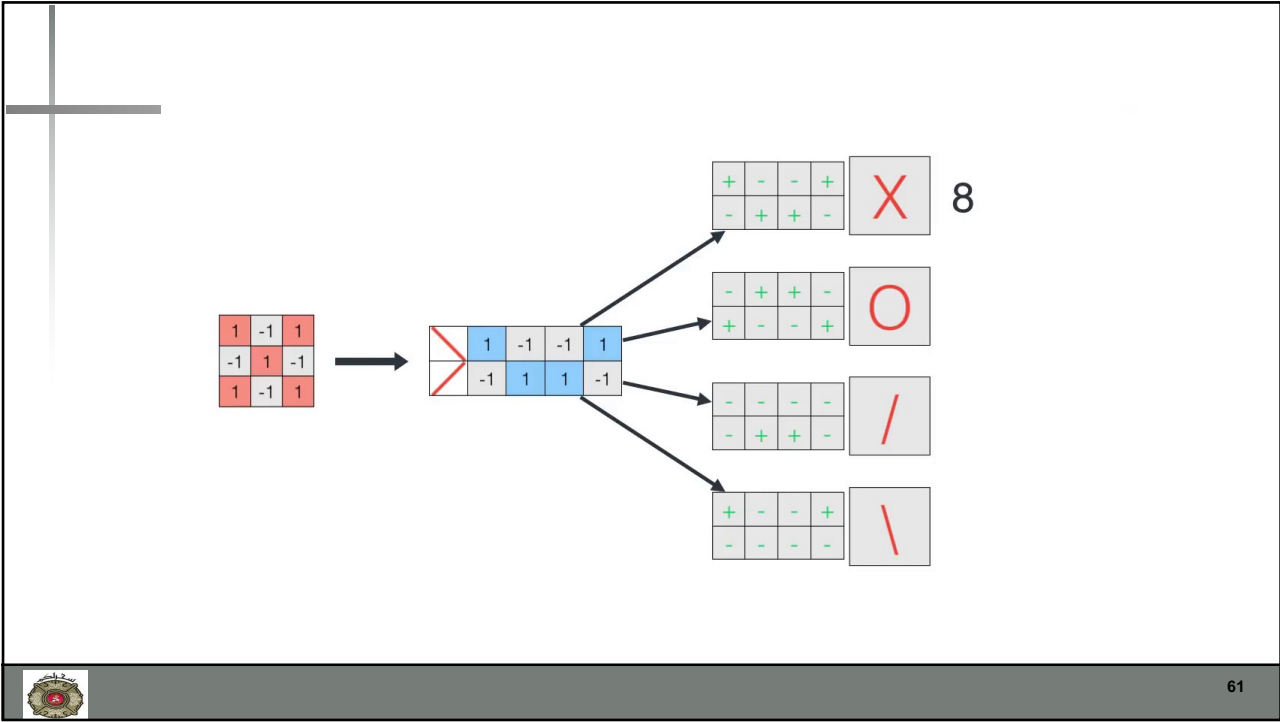
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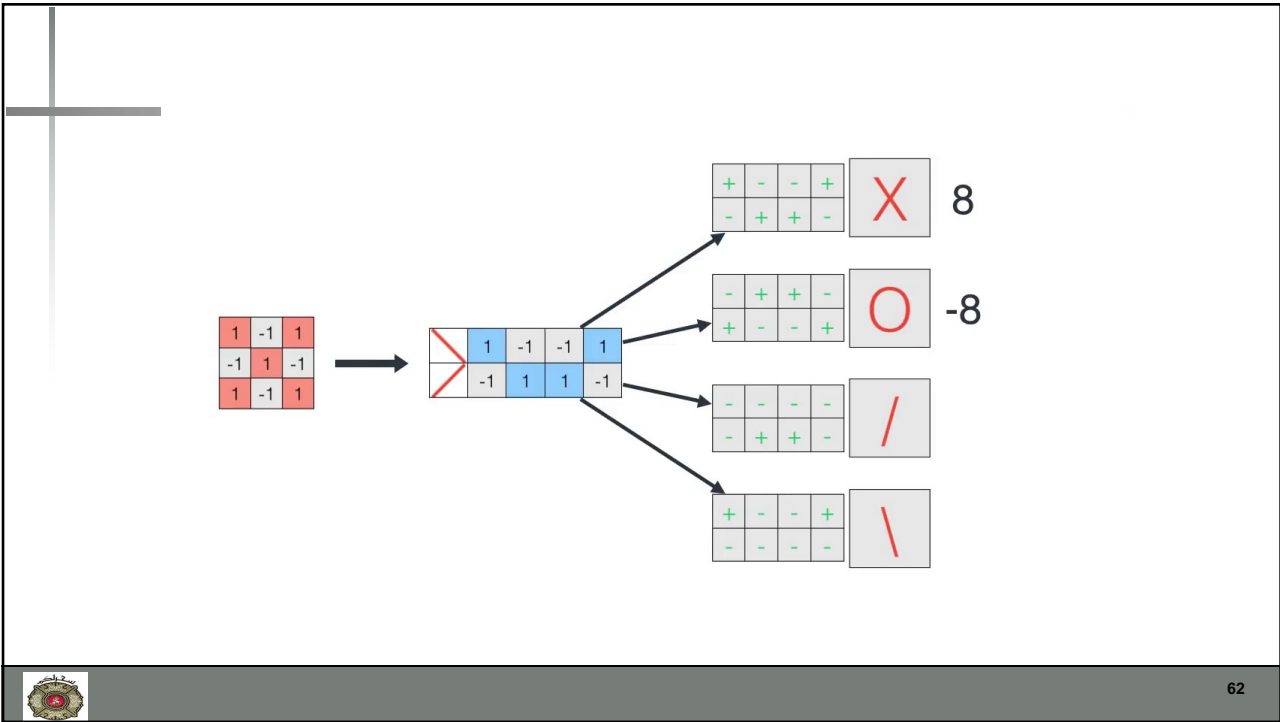
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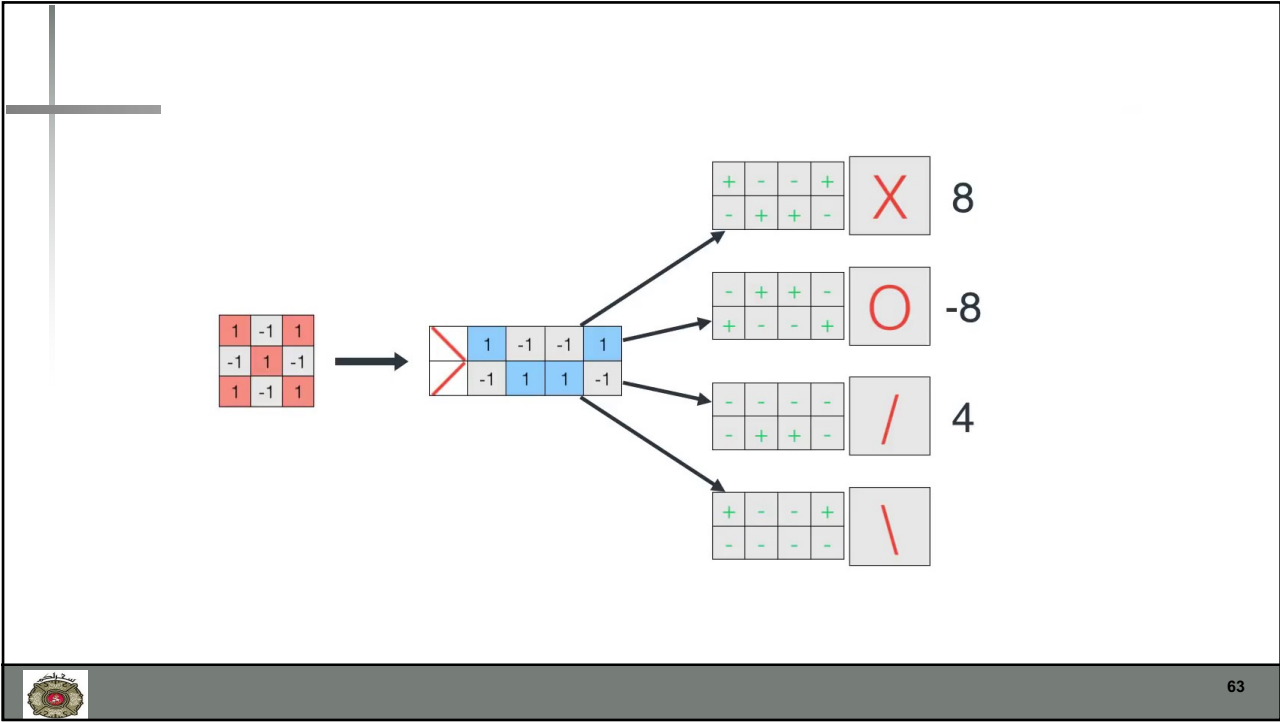
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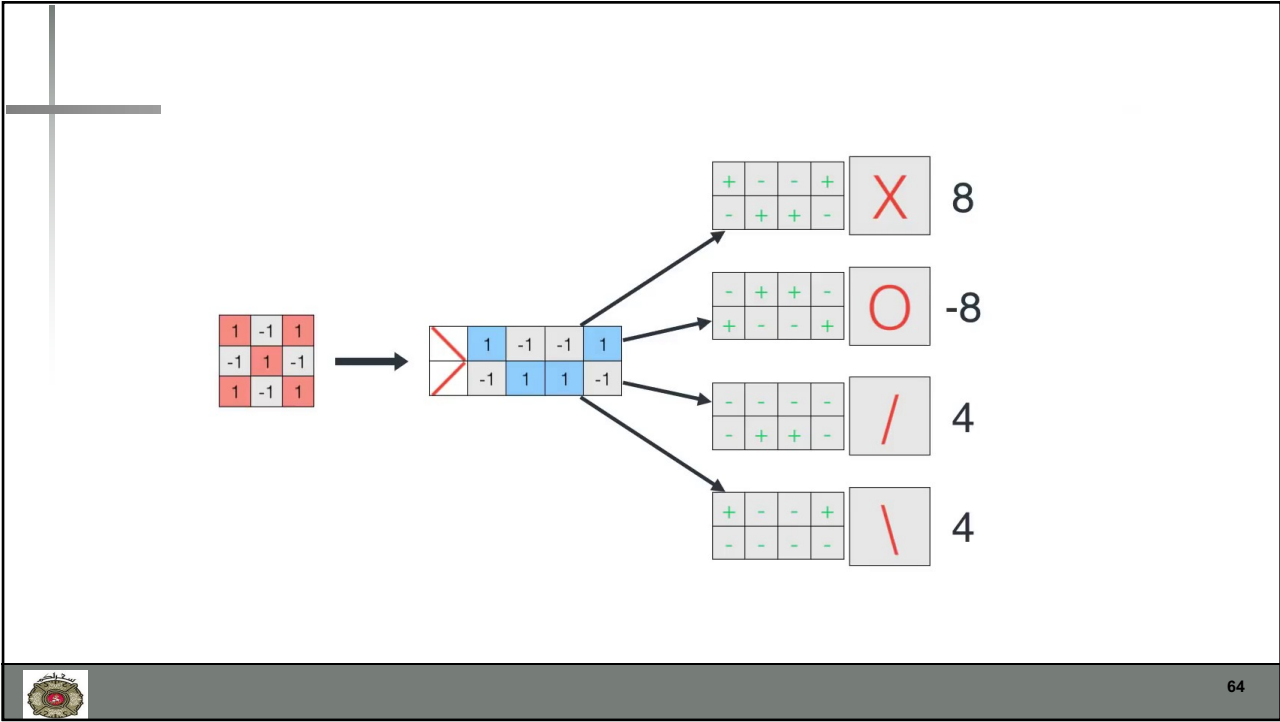
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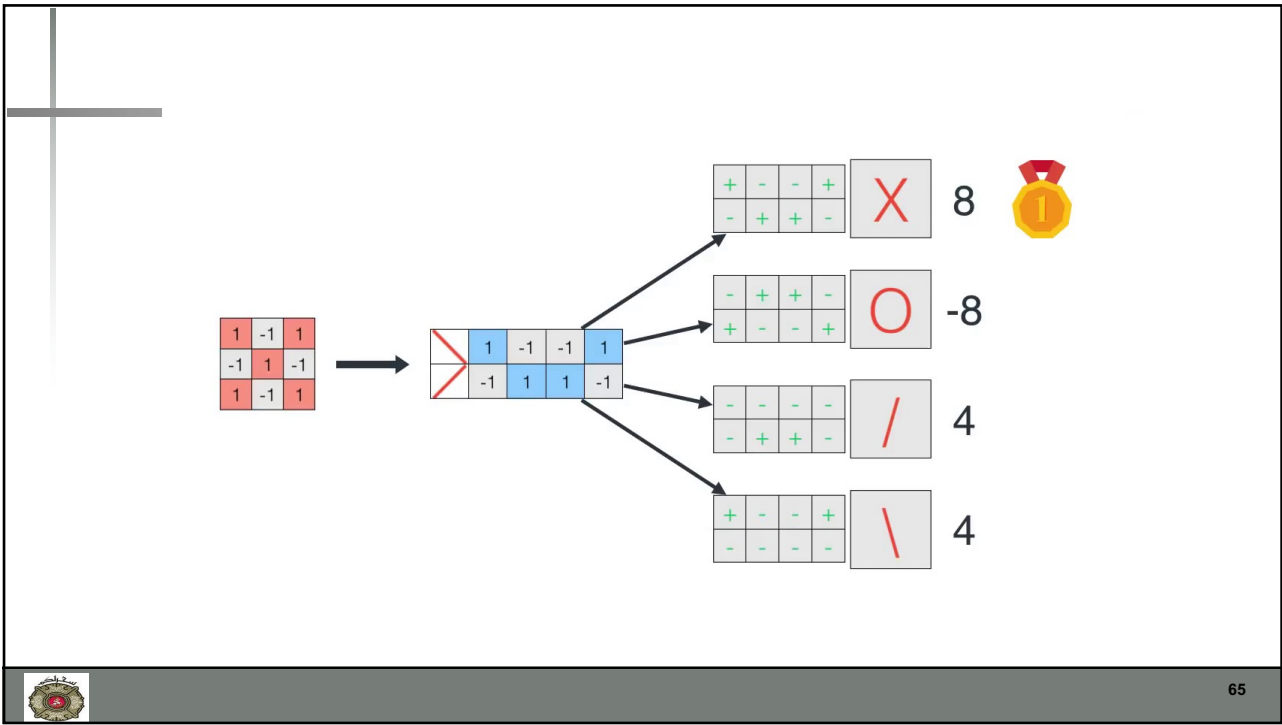
62



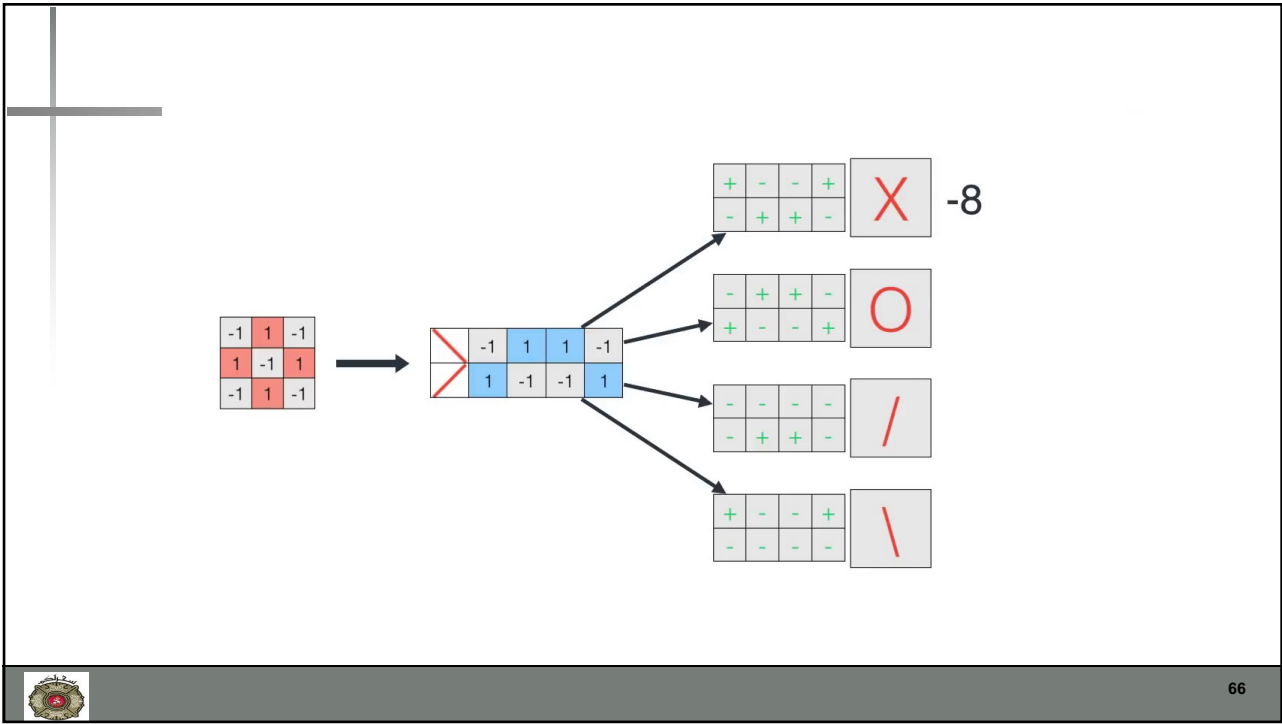
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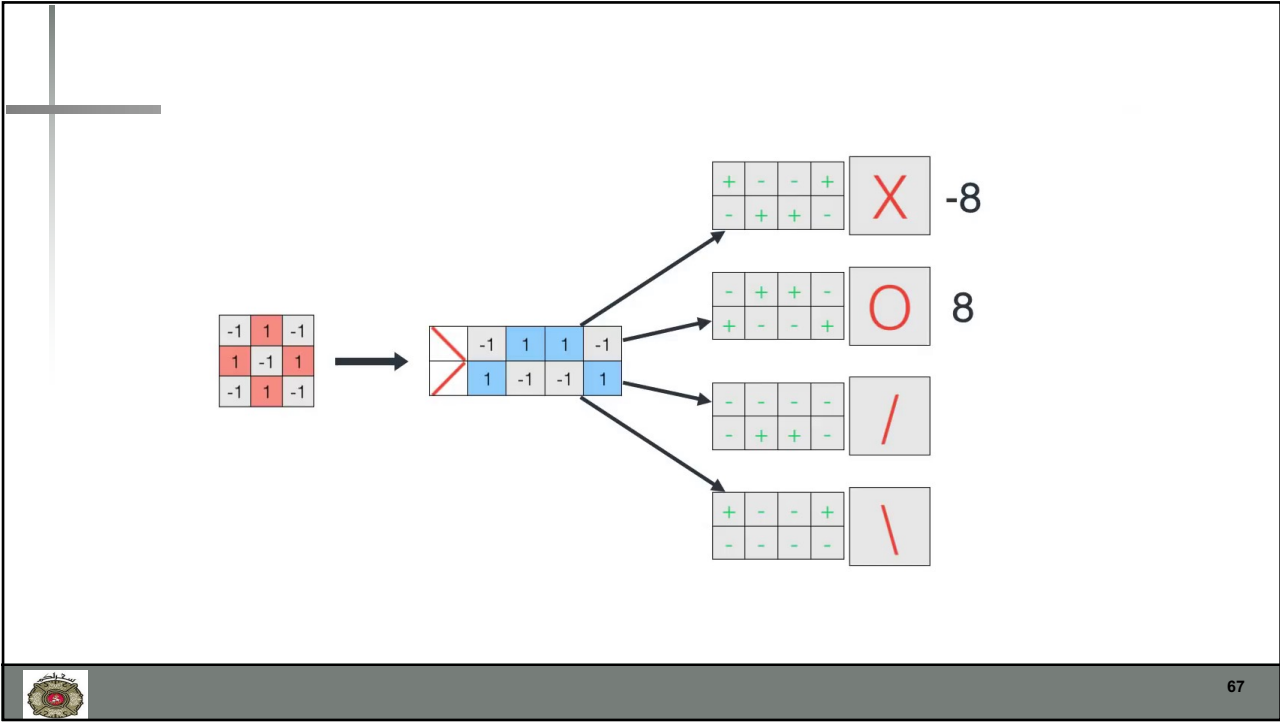
64



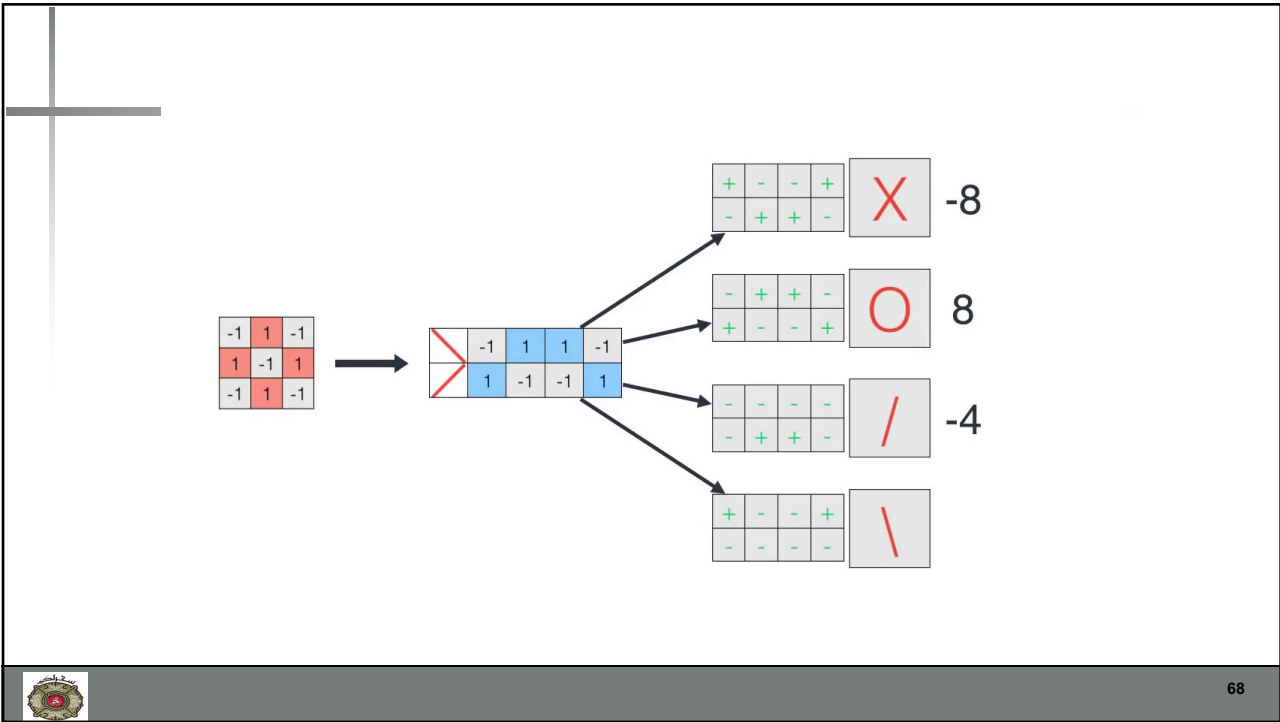
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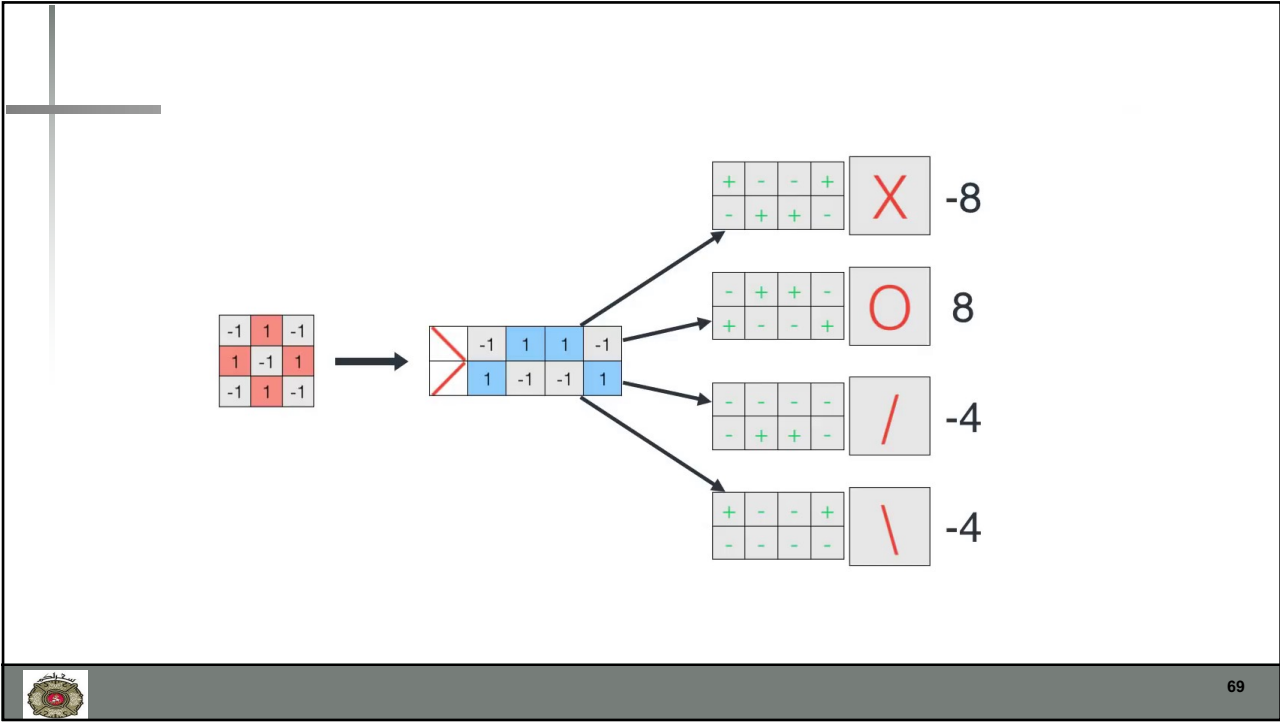
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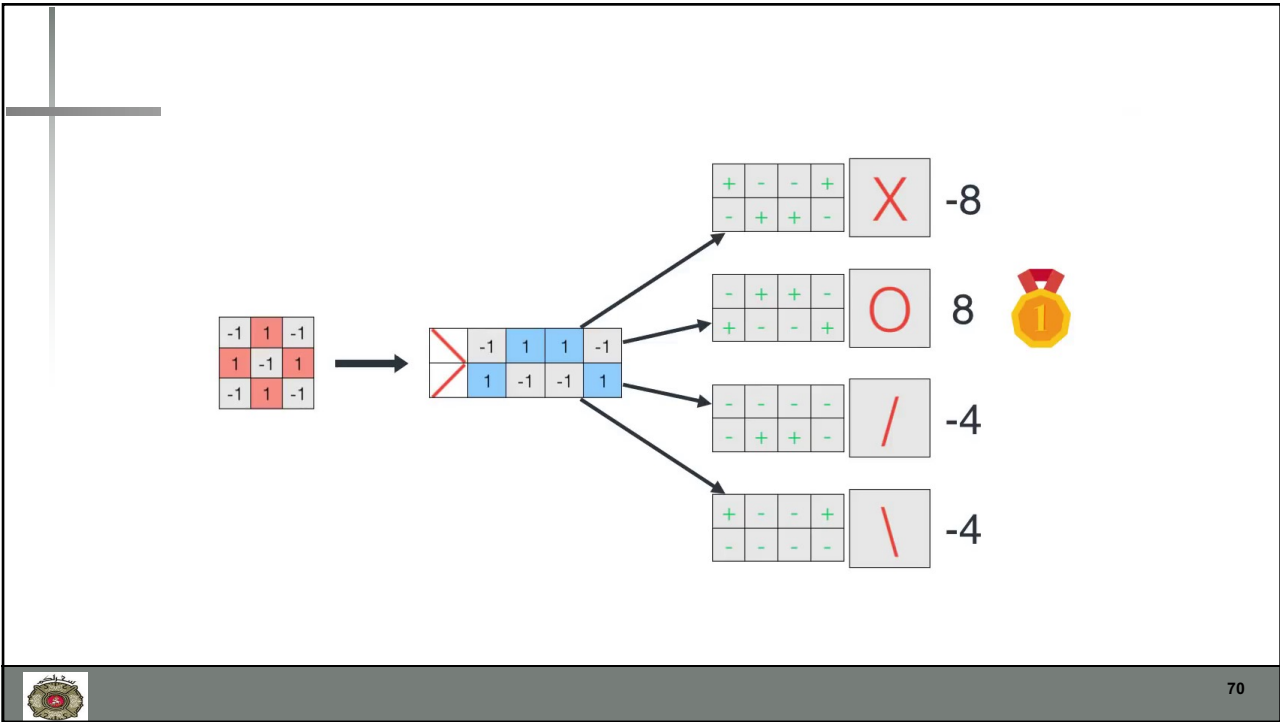
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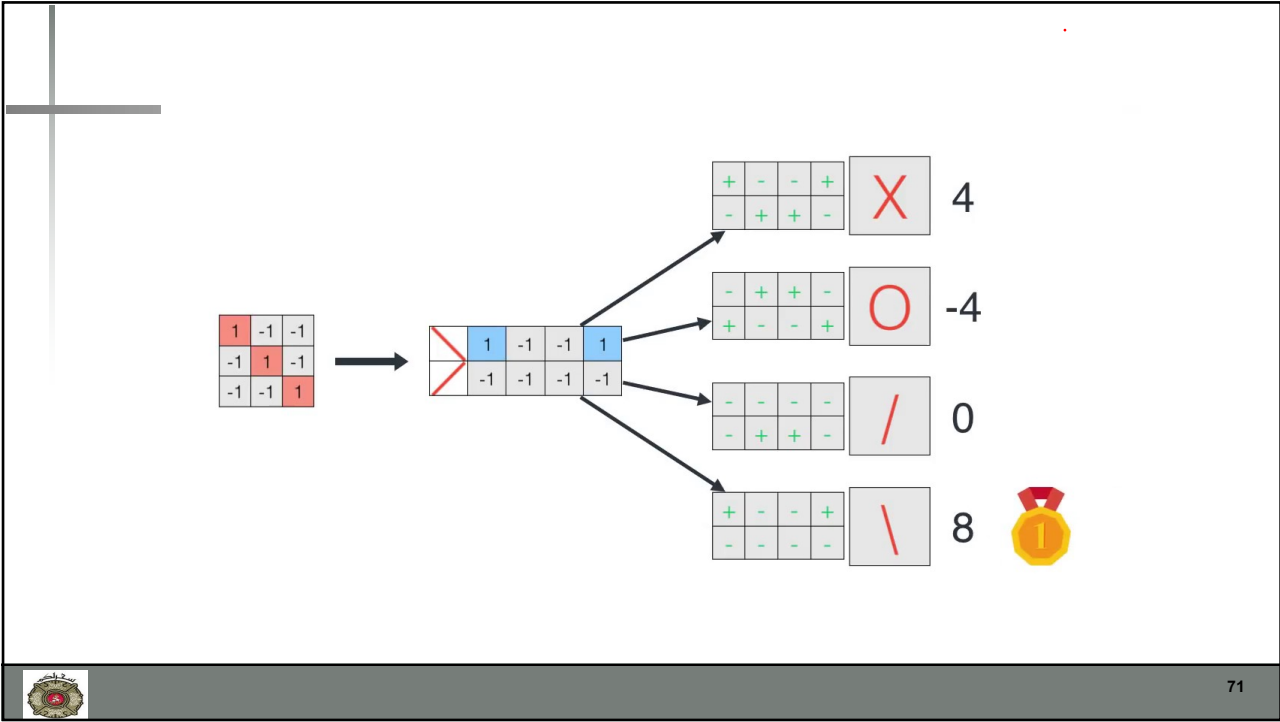
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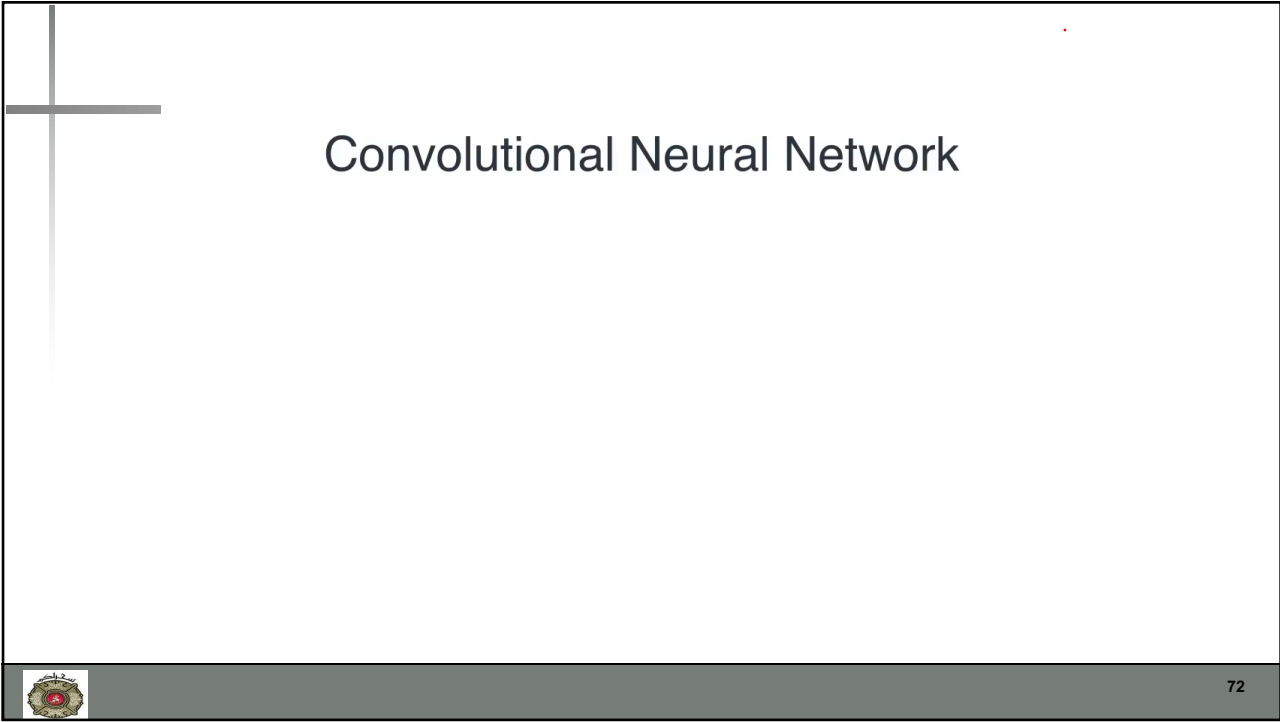
69



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71



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Convolutional Neural Network

1	-1	1
-1	1	-1
1	-1	1

-1	1	-1
1	-1	1
-1	1	-1

-1	-1	1
-1	1	-1
1	-1	-1

1	-1	-1
-1	1	-1
-1	-1	1



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Convolutional Neural Network

1	-1	1
-1	1	-1
1	-1	1

-1	1	-1
1	-1	1
-1	1	-1

-1	-1	1
-1	1	-1
1	-1	-1

1	-1	-1
-1	1	-1
-1	-1	1





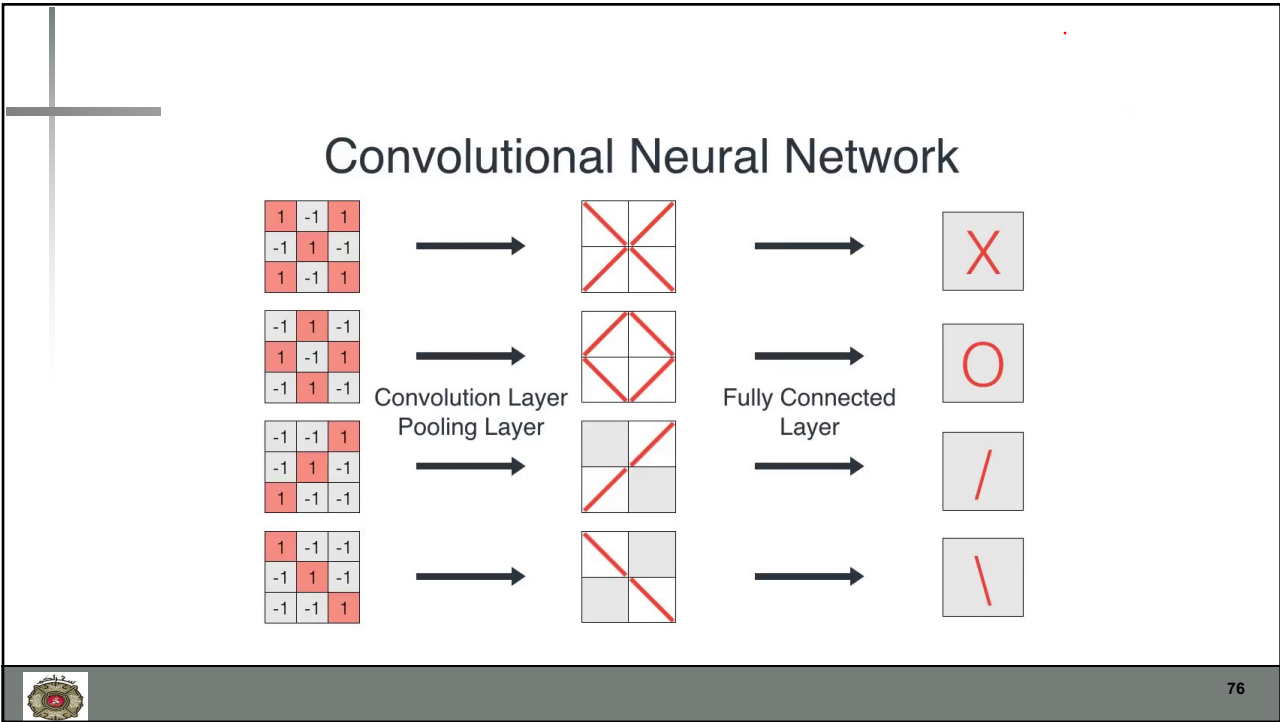
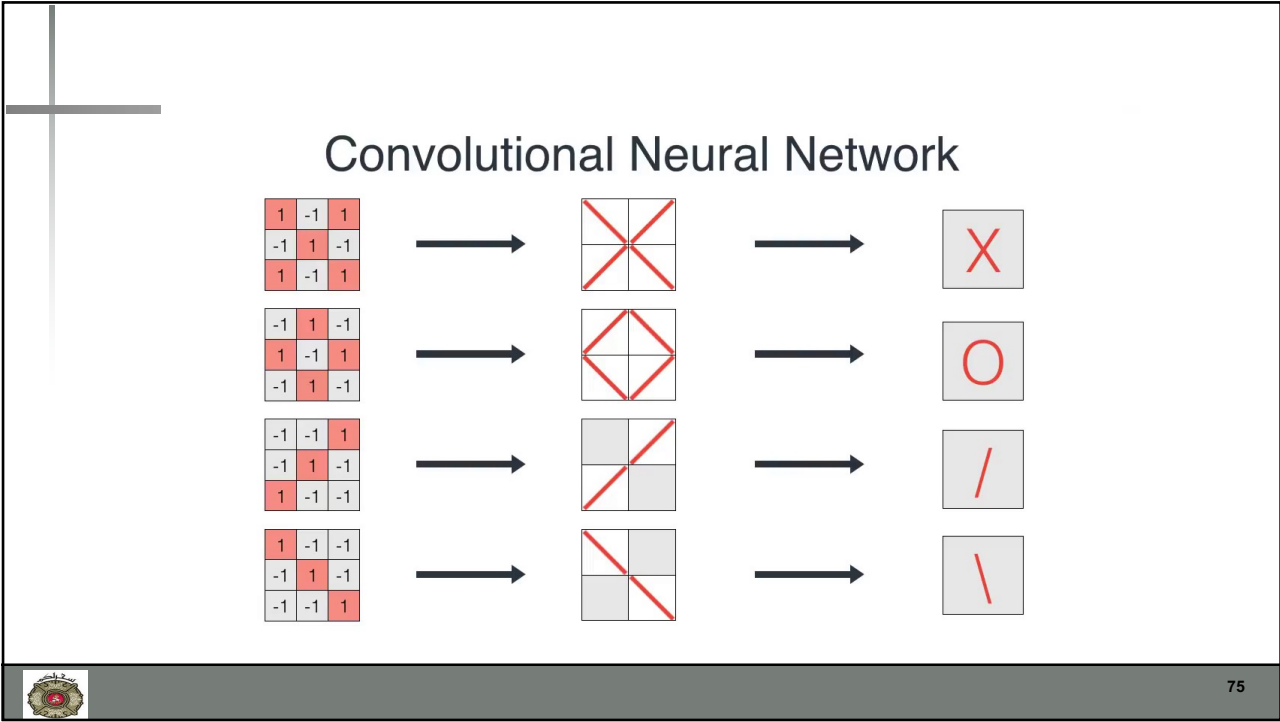






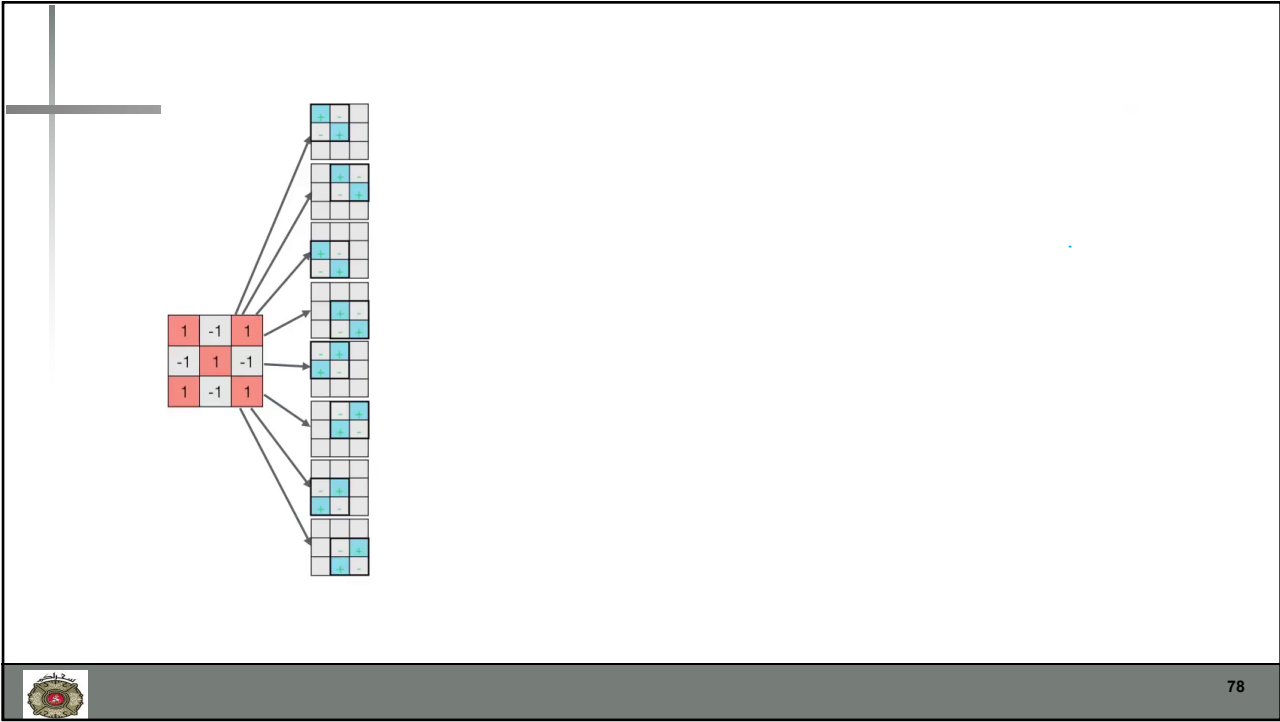
74

74

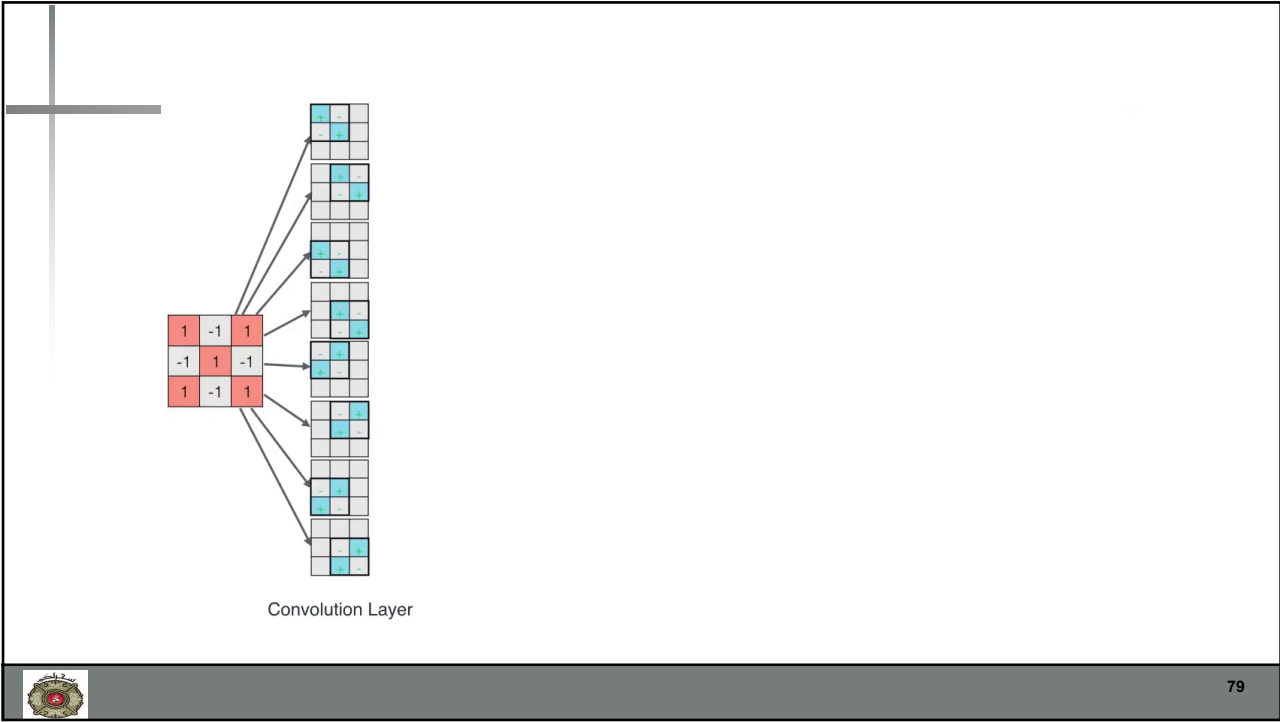




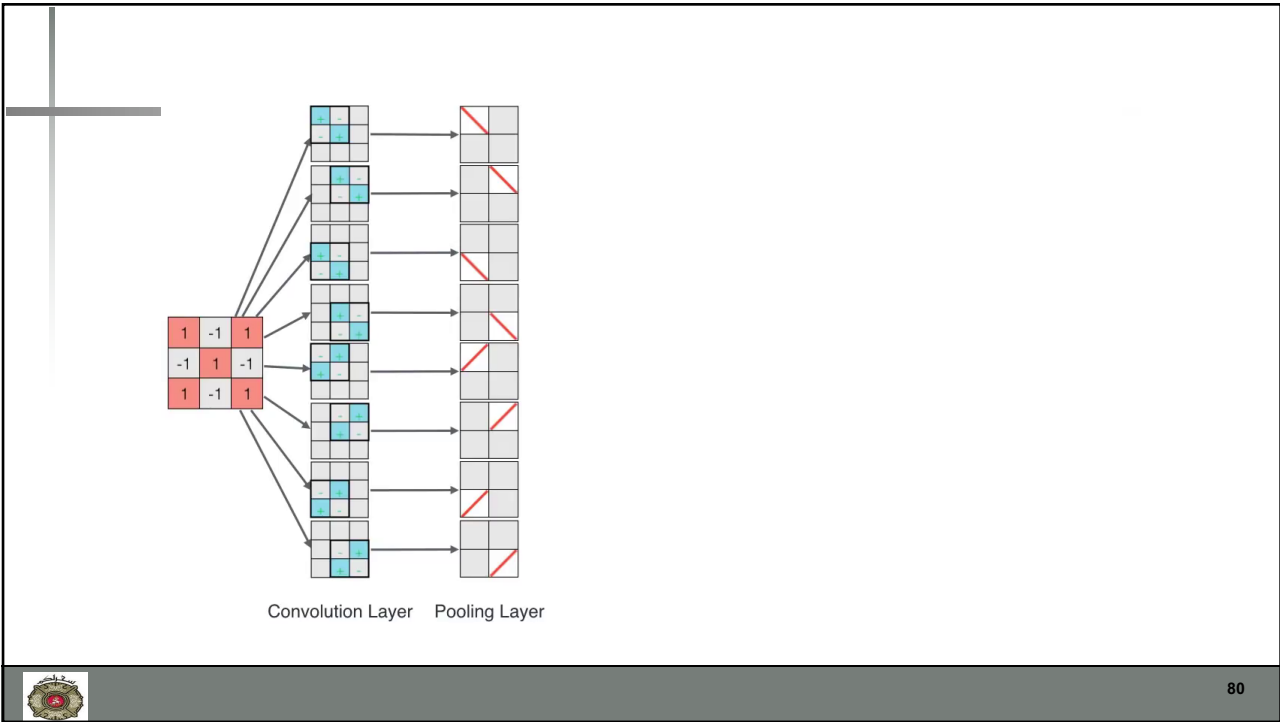
77



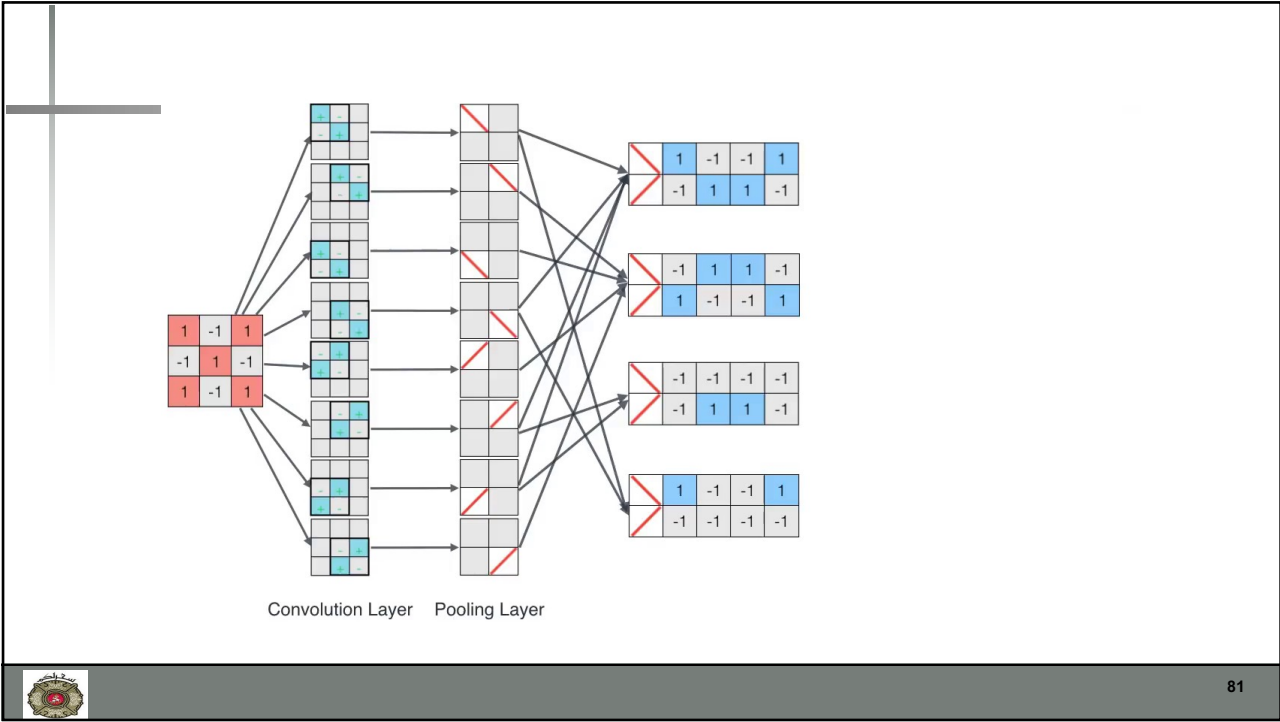
78



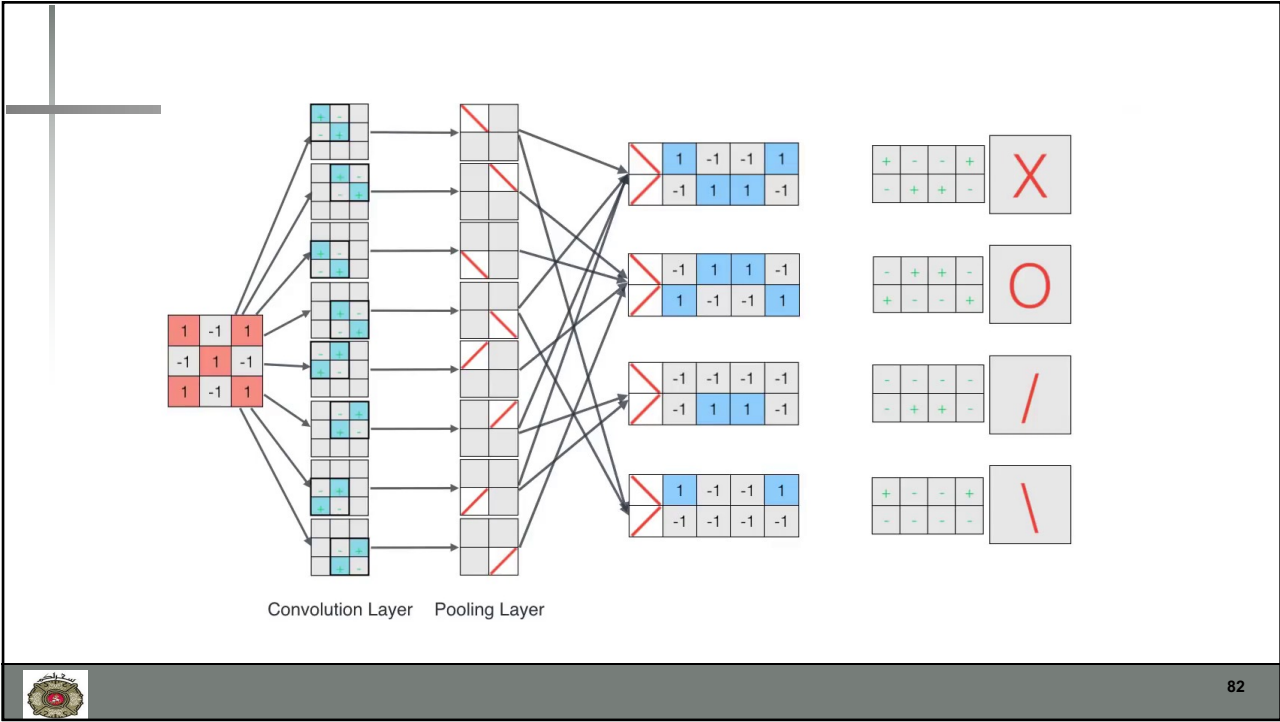
79



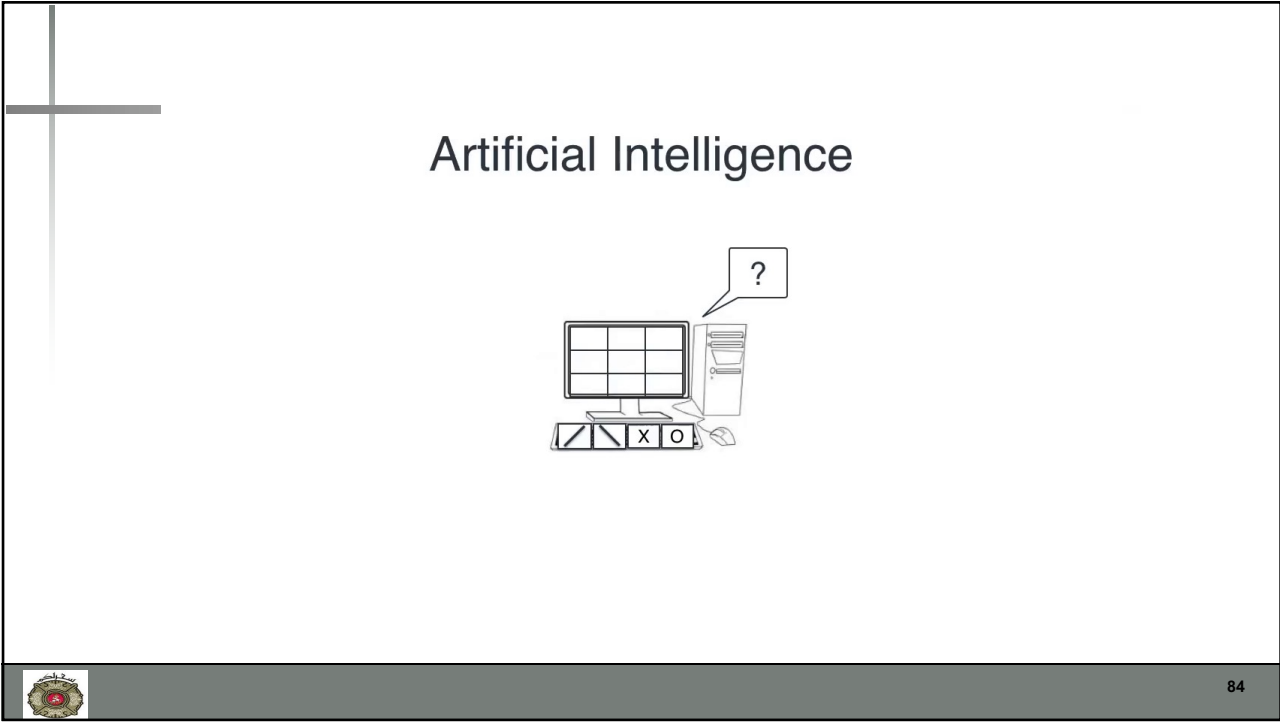
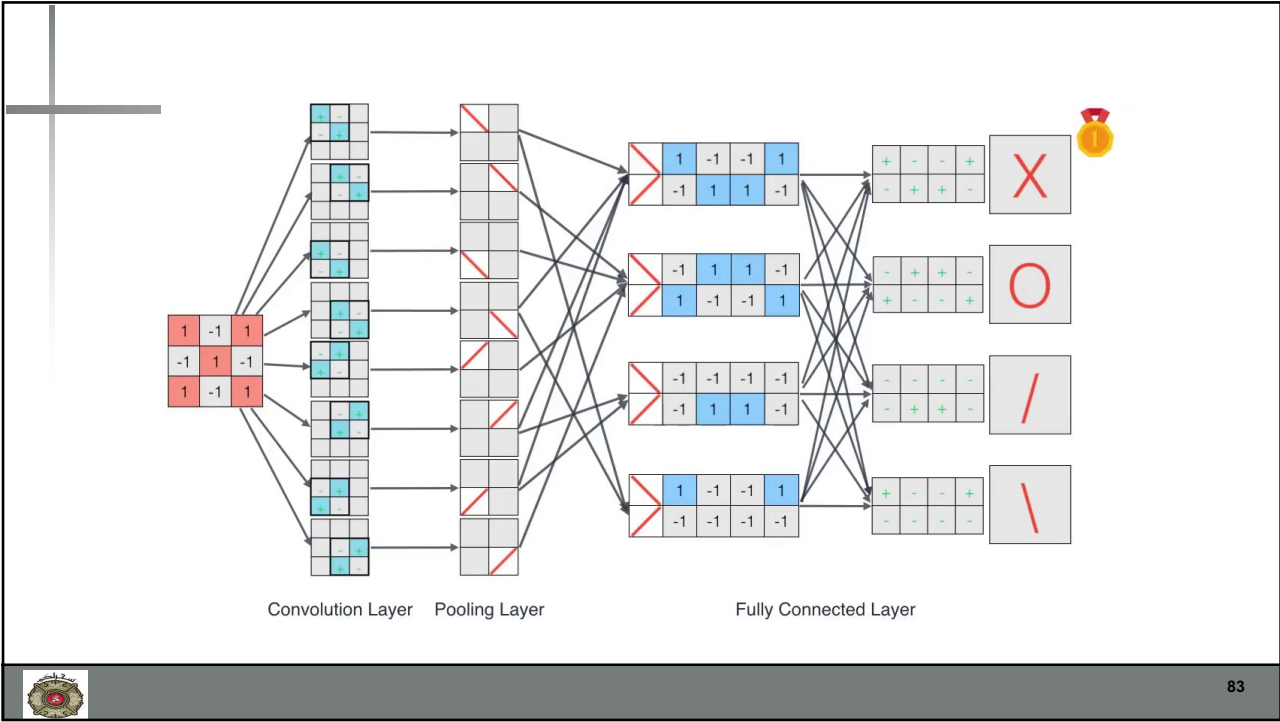
80



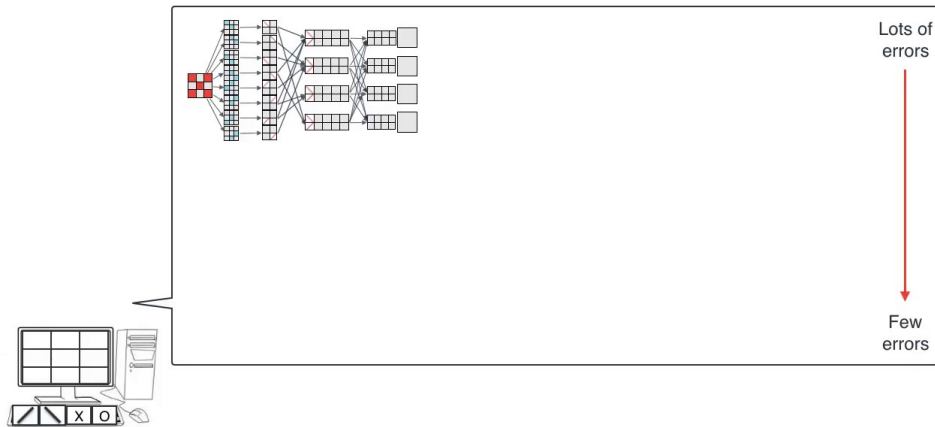
81



82



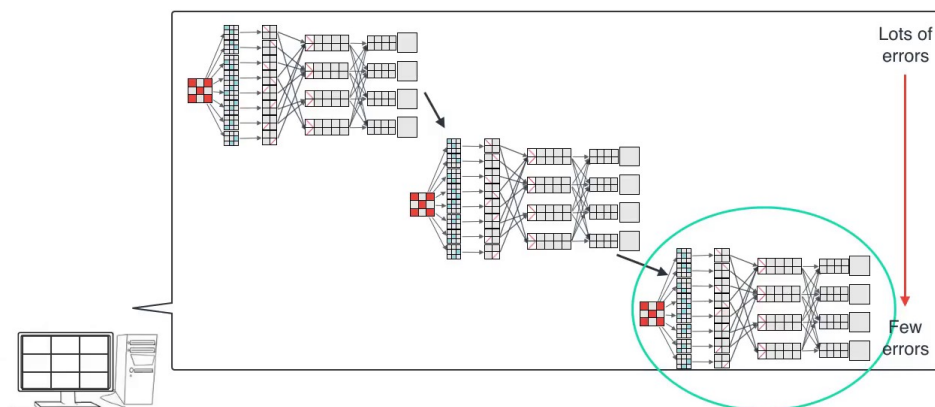
Gradient Descent



85

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Gradient Descent



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Advanced World



People



Computer



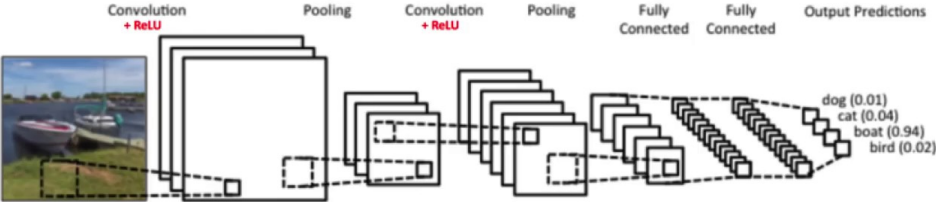
Alphabet



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Convolution + ReLU Pooling Convolution + ReLU Pooling Fully Connected Fully Connected Output Predictions

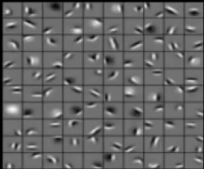




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Deep Learning learns layers of features





Linear Transformation



Linear Transformation



Linear Transformation



Linear Transformation



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Convolutional Neural Networks

- Pooling

Single depth slice

1

1

2

4

5

6

7

8

3

2

1

0

1

2

3

4

x

y

max pool with 2x2 filters and stride 2

6

8

3

4

90

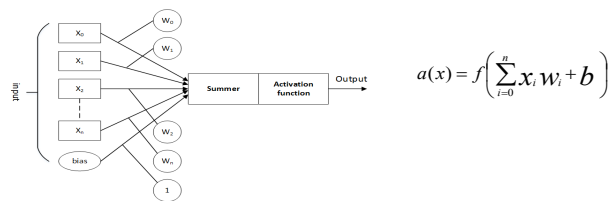
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NUST

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Convolutional Neural Networks

- Activation Functions
 - Building block of an artificial neural network
 - Electronically modeled biological neuron
 - Equipped with some computational strength
 - Multiple input-single output
 - Few hundred to few thousands



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Convolutional Neural Networks (Activation Function)

- Characterize the performance and complexity of the deep neural networks
- Mapping of input data to output data
- Roundedness of the network
- Computations required to calculate the activation function and its derivative dictate the speed of the neural network



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Convolutional Neural Networks (Activation Function)

- Logistic sigmoid unit

- Limits the amplitude of the output of the neuron over the 0 to +1 closed range
- Gradient vanishes as we increase or decrease x

$$f(x) = \frac{1}{1 + e^{-x}}$$

- Hyperbolic tangent unit

- Limits the amplitude of the output of the neuron over the -1 to +1 closed range
- Closely resembles the logistic sigmoid unit

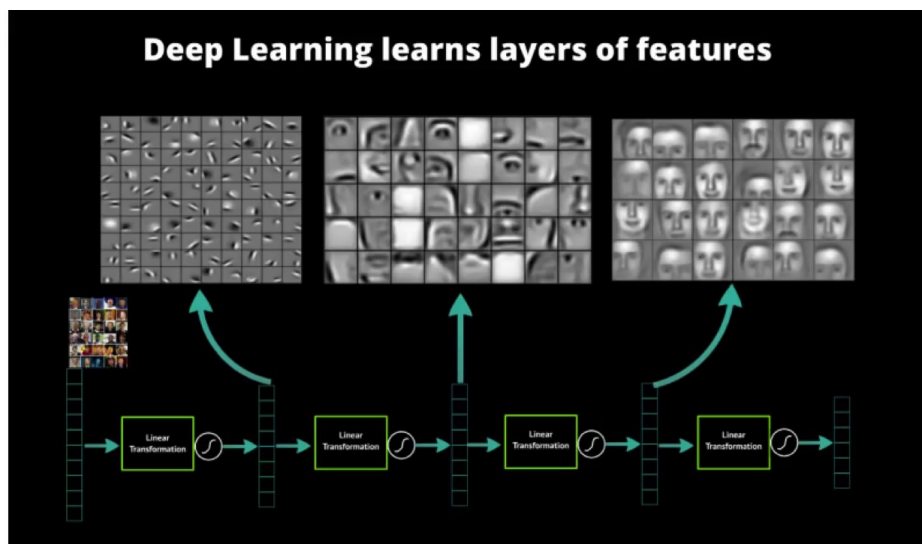
$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



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Deep Learning learns layers of features

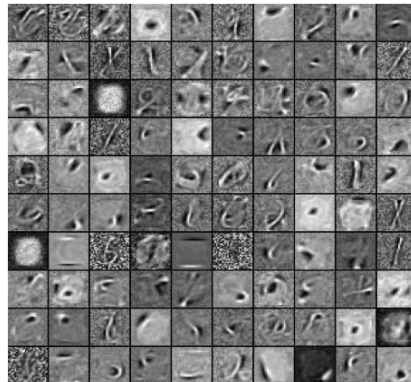


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Convolutional Neural Networks

- Filters learned by the CNN on the MNIST dataset

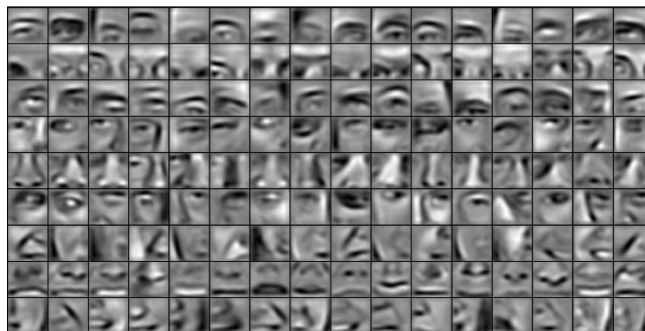


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Convolutional Neural Networks

- Filters learned by the CNN while training on human faces

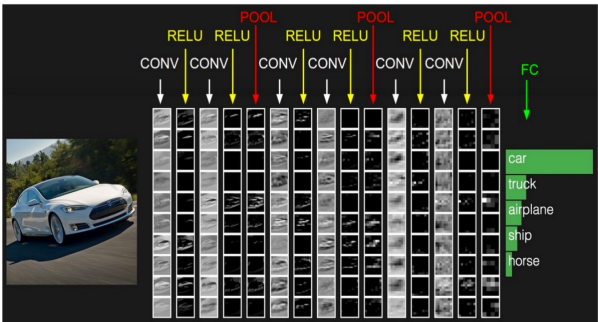


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Convolutional Neural Networks

- Filters learned by the CNN while training on car images



Convolutional Neural Networks

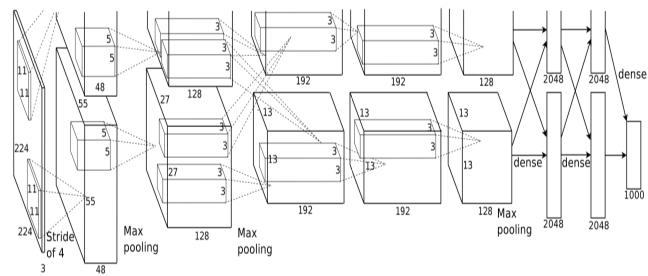
Layer		Feature Map	Size	Kernel Size	Stride	Activation
Input	Image	1	32x32	-	-	-
1	Convolution	6	28x28	5x5	1	tanh
2	Average Pooling	6	14x14	2x2	2	tanh
3	Convolution	16	10x10	5x5	1	tanh
4	Average Pooling	16	5x5	2x2	2	tanh
5	Convolution	120	1x1	5x5	1	tanh
6	FC	-	84	-	-	tanh
Output	FC	-	10	-	-	softmax

LeNet-5 Architecture Summarized Table



AlexNet

■ Architecture



AlexNet

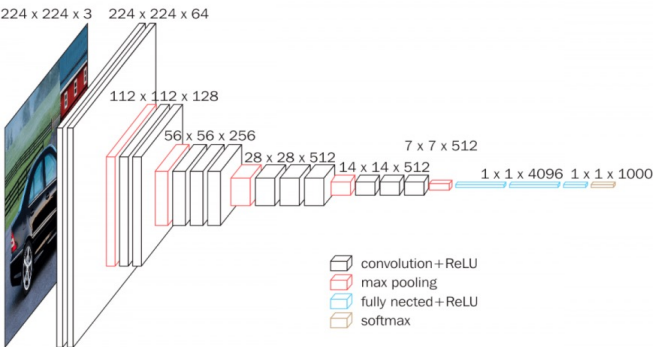
■ Parameters

AlexNet Network - Structural Details													
Input		Output		Layer	Stride	Pad	Kernel size	in	out	# of Param			
227	227	3	55	55	96	conv1	4	0	11	11	3	96	34944
55	55	96	27	27	96	maxpool1	2	0	3	3	96	96	0
27	27	96	27	27	256	conv2	1	2	5	5	96	256	614656
27	27	256	13	13	256	maxpool2	2	0	3	3	256	256	0
13	13	256	13	13	384	conv3	1	1	3	3	256	384	885120
13	13	384	13	13	384	conv4	1	1	3	3	384	384	1327488
13	13	384	13	13	256	conv5	1	1	3	3	384	256	884992
13	13	256	6	6	256	maxpool5	2	0	3	3	256	256	0
							fc6		1	1	9216	4096	37752832
							fc7		1	1	4096	4096	16781312
							fc8		1	1	4096	1000	4097000
Total										62,378,344			



VGGNet (VGG16)

■ Architecture

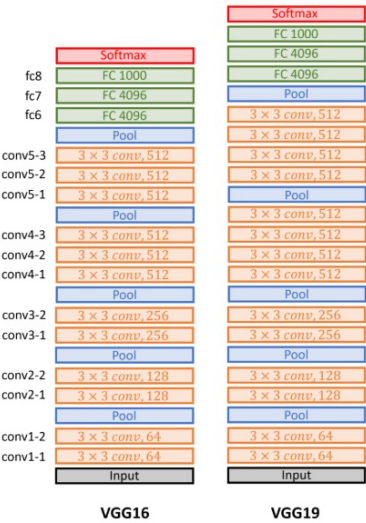


VGGNet (VGG16)

VGG16 - Structural Details													
#	Input Image			output			Layer	Stride	Kernel	in	out	Param	
1	224	224	3	224	224	64	conv3-64	1	3	3	3	64	1792
2	224	224	64	224	224	64	conv3-64	1	3	3	64	64	36928
3	224	224	64	112	112	64	maxpool	2	2	2	64	64	0
4	112	112	64	112	112	128	conv3-128	1	3	3	64	128	73856
5	112	112	128	112	112	128	conv3-128	1	3	3	128	128	147584
6	112	112	128	56	56	128	maxpool	2	2	2	128	128	65664
7	56	56	128	56	56	256	conv3-256	1	3	3	128	256	295168
8	56	56	256	56	56	256	conv3-256	1	3	3	256	256	590080
9	56	56	256	56	56	256	conv3-256	1	3	3	256	256	590080
10	56	56	256	28	28	256	maxpool	2	2	2	256	256	0
11	28	28	256	28	28	512	conv3-512	1	3	3	256	512	1180160
12	28	28	512	28	28	512	conv3-512	1	3	3	512	512	2359808
13	28	28	512	28	28	512	conv3-512	1	3	3	512	512	2359808
14	28	28	512	14	14	512	maxpool	2	2	2	512	512	0
15	14	14	512	14	14	512	conv3-512	1	3	3	512	512	2359808
16	14	14	512	14	14	512	conv3-512	1	3	3	512	512	2359808
17	14	14	512	7	7	512	maxpool	2	2	2	512	512	0
18	1	1	25088	1	1	4096	fc		1	1	25088	4096	102764544
19	1	1	4096	1	1	4096	fc		1	1	4096	4096	16781312
20	1	1	4096	1	1	1000	fc		1	1	4096	1000	4097000
Total												138,423,208	

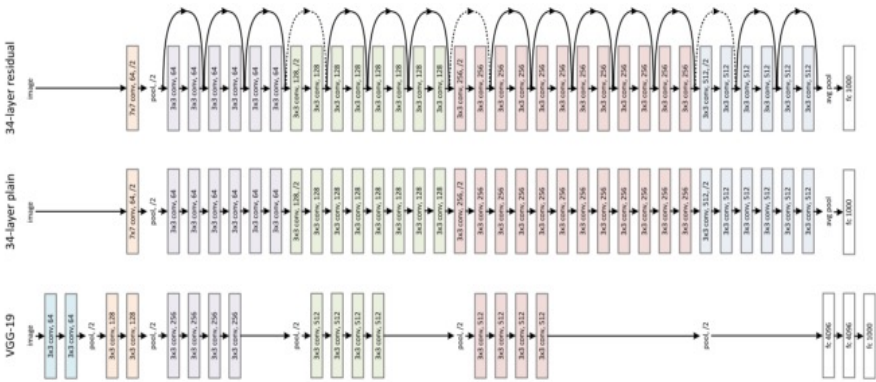


VGGNet (VGG19)



ResNet

■ Architecture



GoogLeNet

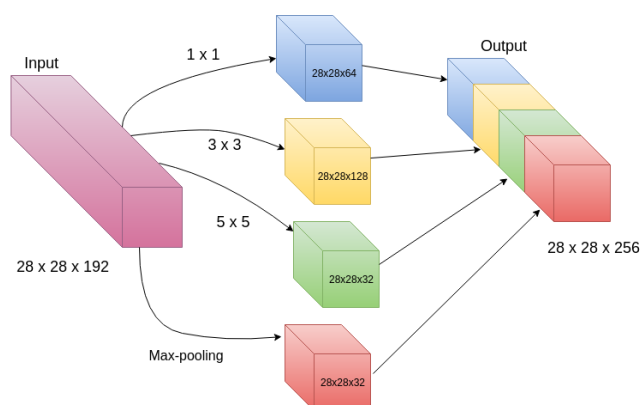
■ Motivation



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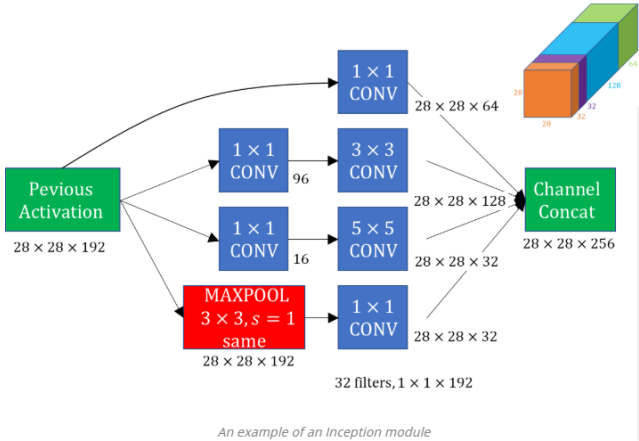
GoogLeNet



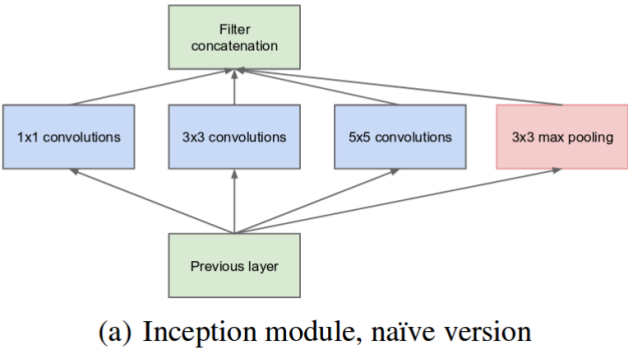
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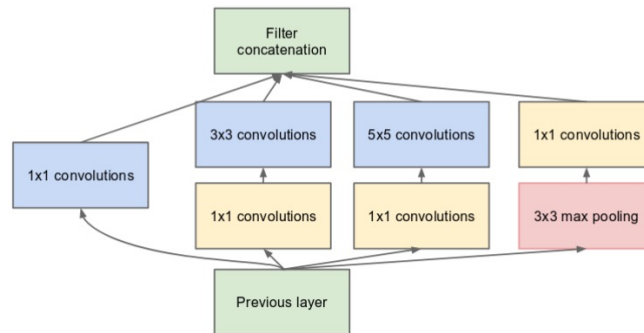
Inception Module



Inception Module



Inception Module



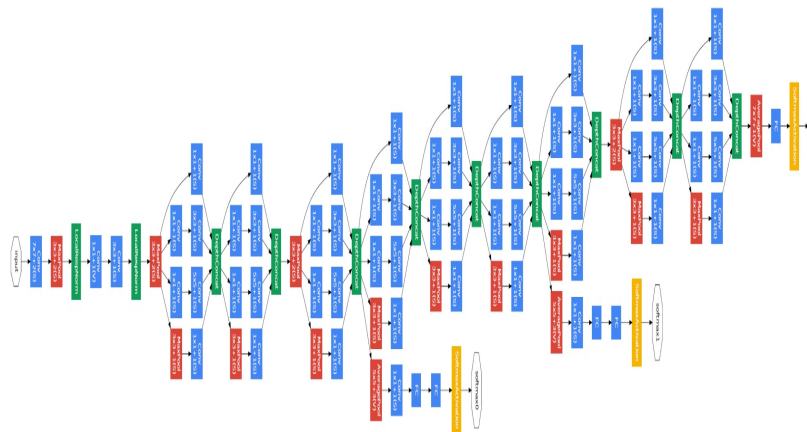
(b) Inception module with dimension reductions



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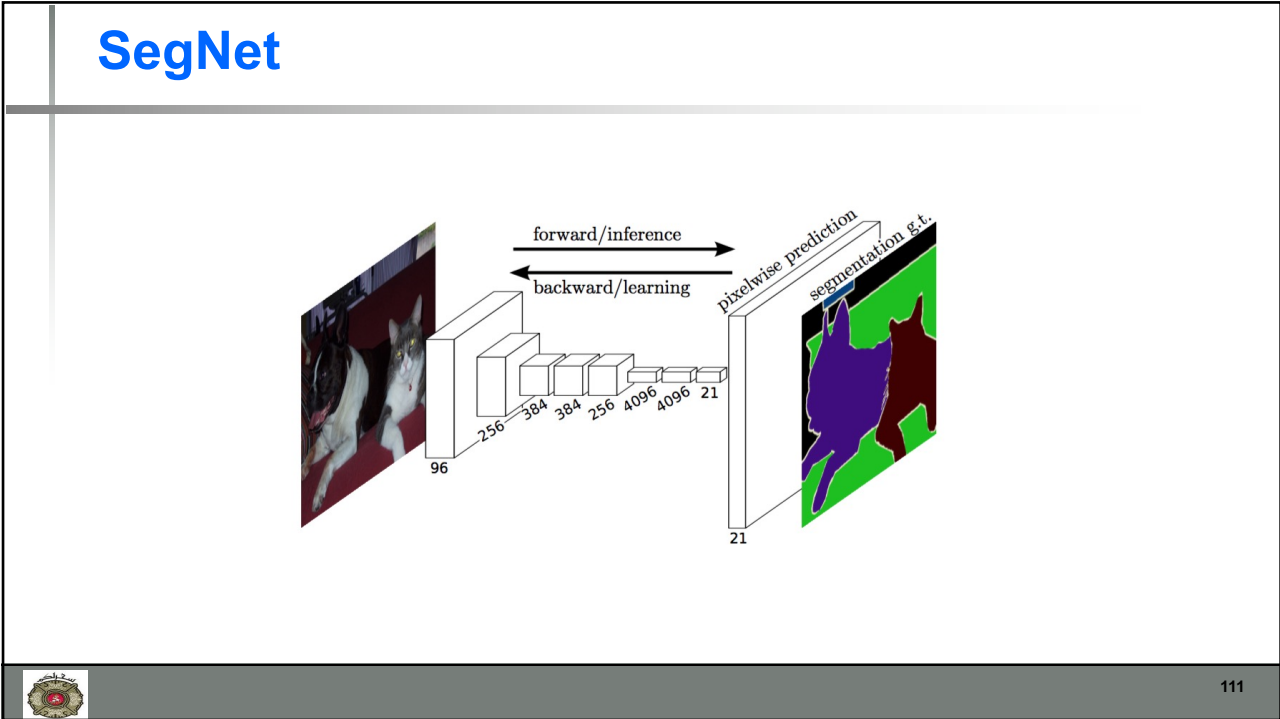
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GoogLeNet

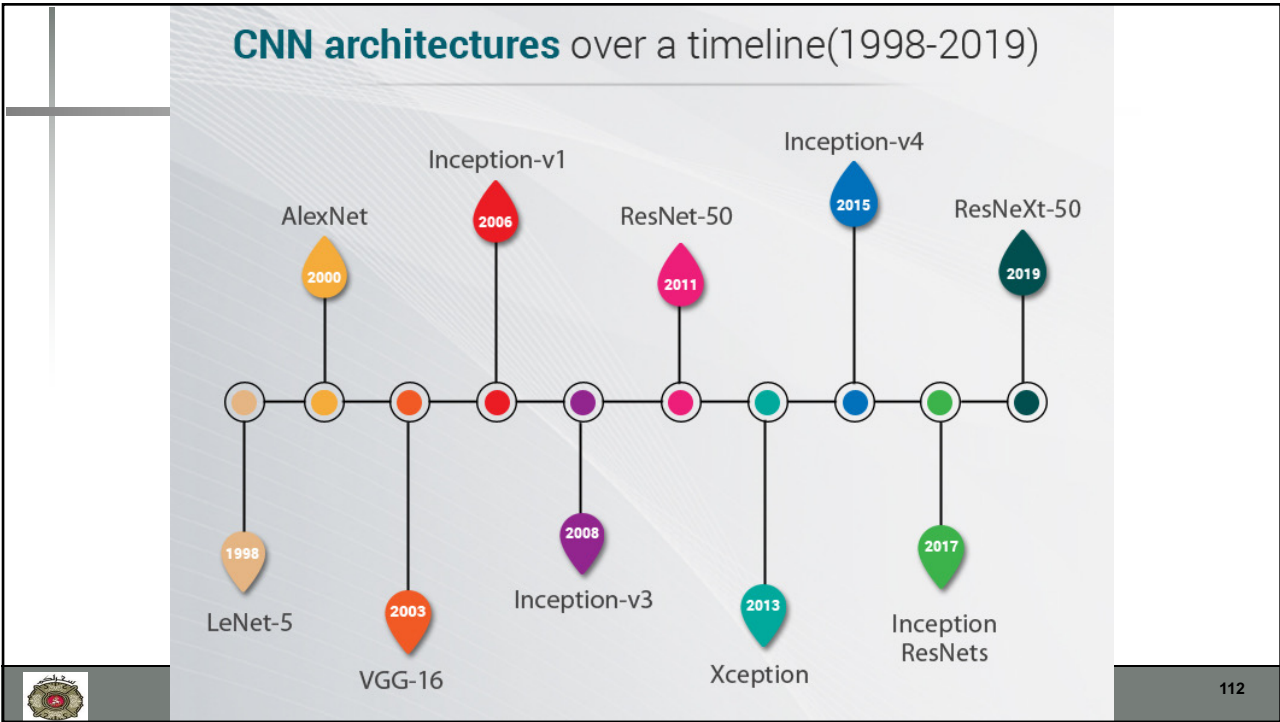


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TRANSFER LEARNING



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Transfer Learning (1)



Transfer learning deals with the ability of a system to recognize and apply knowledge and skills learned in previous tasks/domains to a novel task/domain, which share some commonality



Transfer learning occurs when the learner recognizes features among concepts, skills or principles.



Transfer learning assesses the value of utilizing what was learnt in one situation in another



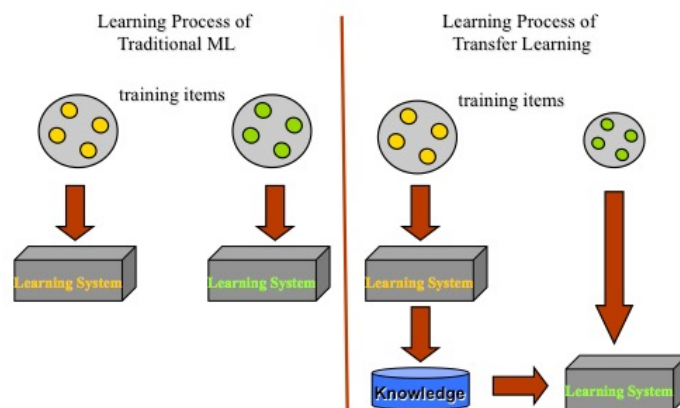
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Transfer Learning (2)

Traditional ML vs. TL



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Transfer Learning (3)

- Examples of transfer learning
 - Knowledge of French may help students learn Spanish
 - Learning to drive a car helps a person to later drive a truck
 - Learning mathematics prepares students to learn physics



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Reading Material



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Thank you



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